Module 2

SCALAR & VECTOR POINTS

Scalar and point techniques – vector visualization techniques – matrix visualization, Visualization Techniques for Trees, Graphs, and Networks, Multidimensional data, Visual Variables- Networks and Trees - Map Color and Other Channels- Manipulate View- Heat Map.

Outline for class

- ✓ Scalar and Point techniques
- ✓ Color maps
- ✓ Contouring – Height Plots - Vector visualization techniques
- ✓ Vector properties – Vector Glyphs – Vector Color Coding
- ✓ Matrix visualization techniques

Scalar Visualisation

- Scalar is a physical quantity that contains only magnitude.
- Visualizing scalar data is frequently encountered in science, engineering, and medicine, but also in daily life.
- There exist many scalar visualization techniques, both for 2D and 3D datasets.
- The most popular scalar visualization techniques:
  - Color mapping
  - Contouring
  - Heatmaps
  - Topological Analysis and height plots.

Scalar Visualization

Color Mapping

- Color mapping is a common scalar visualization technique that maps scalar data to colors, and displays the colors on the computer system.
- The scalar mapping is implemented by indexing into a *color lookup table*.
- Scalar values then serve as indices into this lookup table.
- The lookup table holds an array of colors.
- Associated with the table is a minimum and maximum scalar range into which the scalars are mapped.
- Scalar values greater than the maximum are clamped to the maximum color, scalar values less than the minimum are clamped to the minimum value.

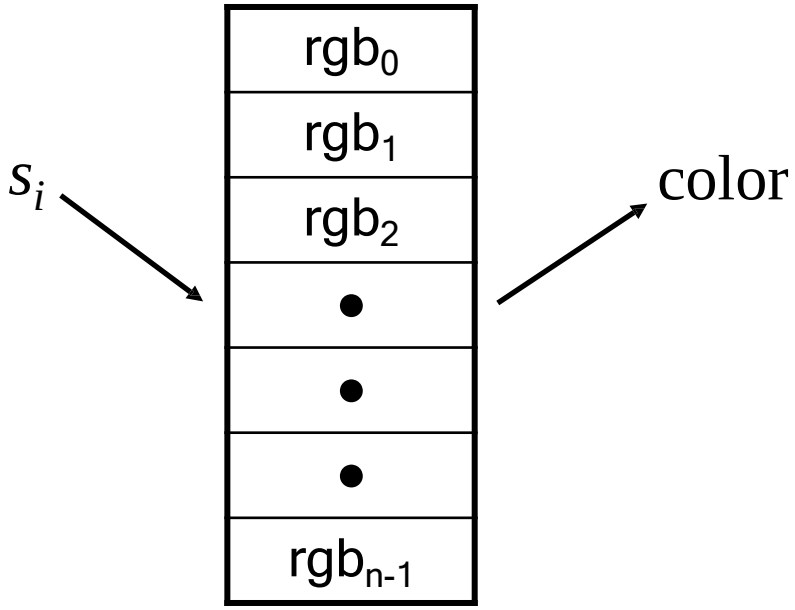
Scalar Visualization

Then, for each scalar value s_i, the index i into the color table with n entries is given as:

$$s_i < \min \quad : \quad i = 0$$

$$s_i > \max \quad : \quad i = n - 1$$

otherwise: $i = n \cdot \left(\dfrac{s_i - \min}{\max - \min} \right)$



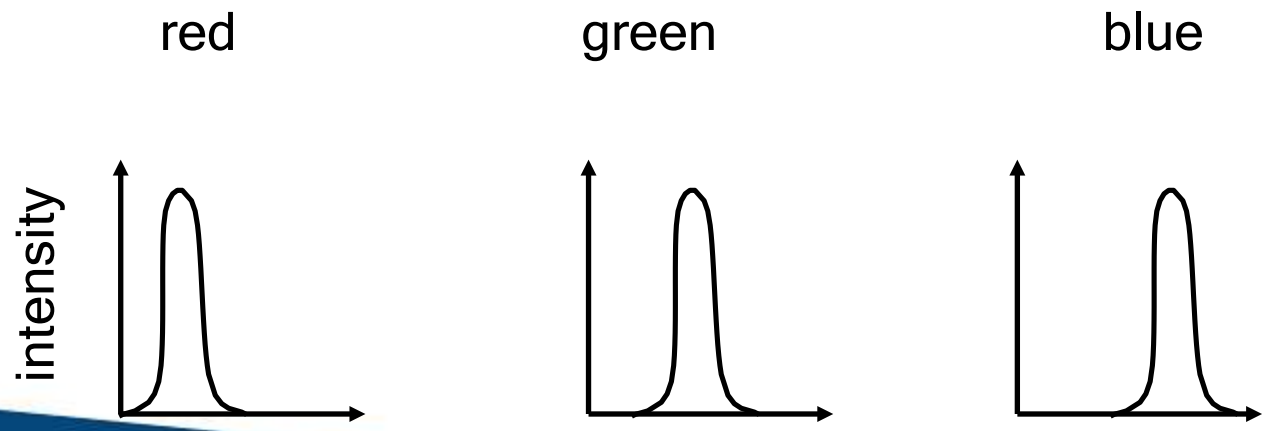
PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

- **si**: This is the scalar value that you want to map to a color.
- **s_min**: This represents the minimum scalar value in your dataset. It defines the lower bound of the scalar range.
- **s_max**: This represents the maximum scalar value in your dataset. It defines the upper bound of the scalar range.
- **n**: This is the number of entries in your color table.

Scalar Visualization

Transfer Functions

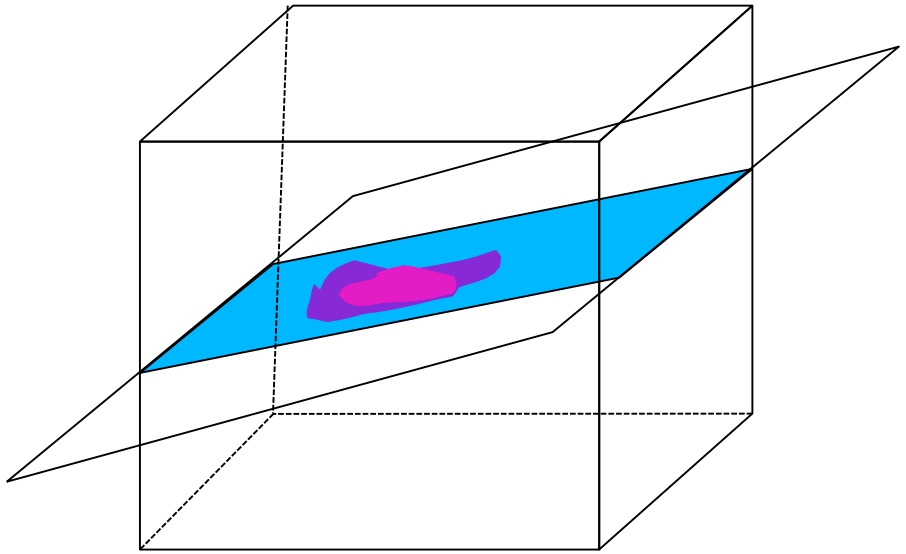
A more general form of the lookup table is called *transfer function*. A transfer function is any expression that maps scalar values into a color specification. For example, a function can be used to map scalar values into separate intensity values for the red, green, and blue components.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

Scalar Visualization

Color mapping is a one-dimensional visualization technique. It maps one piece of information (i.e. a scalar value) into a color specification. However, the display of color information is not limited to one-dimensional displays. Often we use color information mapped onto
1-D, 2-D, or 3-D objects.
This is a simple way to increase the information content of our visualization.
In 3-D, cutting planes can be used to visualize the data inside.

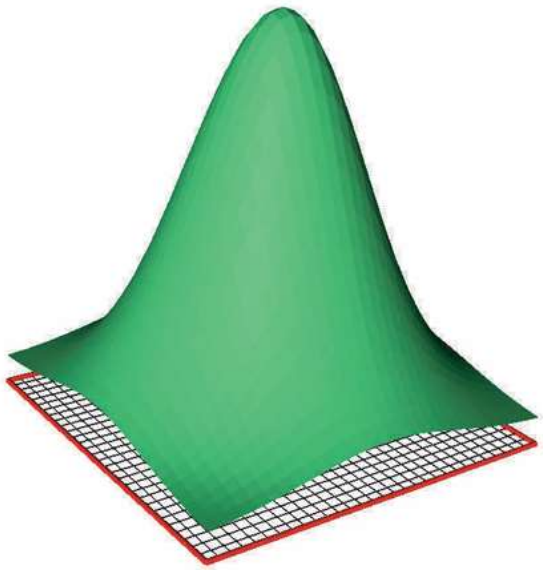


PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Scalar Visualization

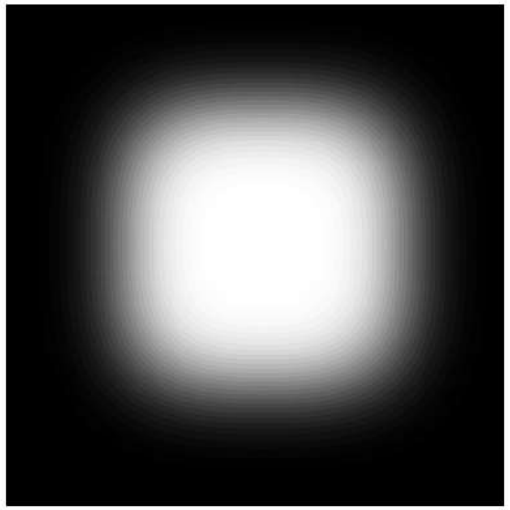
- Many other color map designs are possible. For example, geographical applications often encode landscape height using a particular color map that contains colors, which suggest typical relief forms at different heights, including blue (sea level), green (fields), beige (hills), brown (mountains), and white (mountain peaks).
- In some applications, we want to emphasize the variations of the data rather than absolute data values. This is useful when we are interested in detecting the dataset regions where data changes most quickly or stays constant. For this goal, we can use a color map containing two or more alternating colors that are perceptually very different. When the data values change, the colors change abruptly, yielding the easily detectable band-like patterns in the visualization.

- Visualize a scalar function :
- $f(x,y)=e^{-10(x4+y4)}$, whose shape is quite similar to the Gaussian

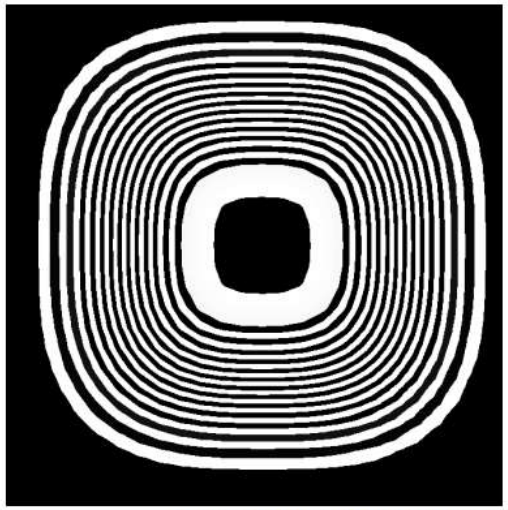


Scalar Visualization

Example:



luminance color map



Images courtesy of Alexandru Telea

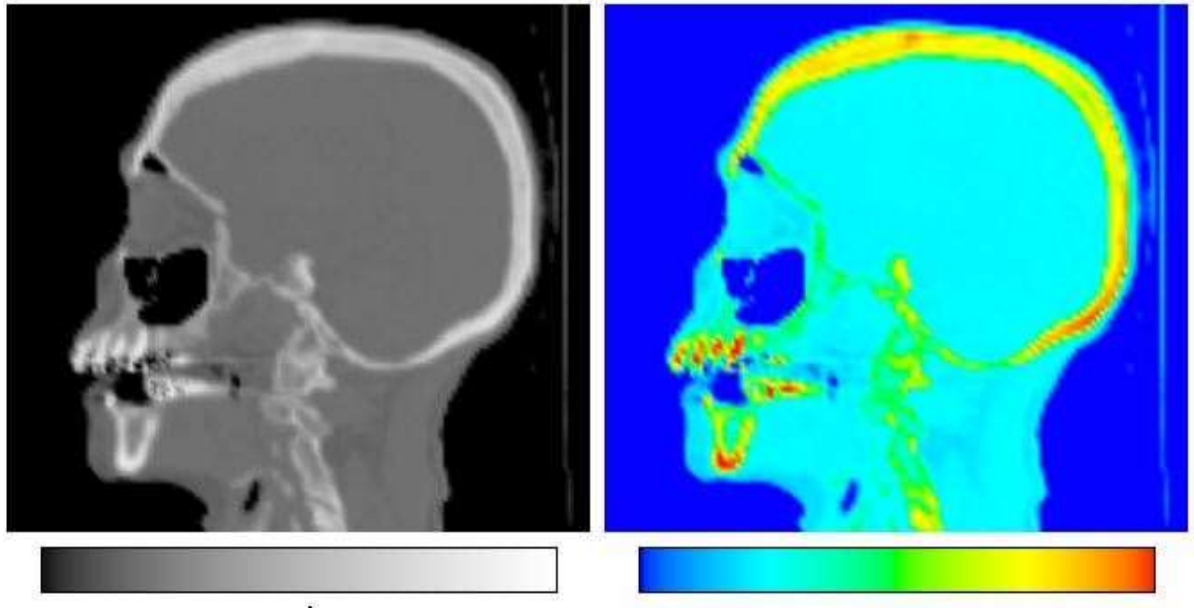
zebra color map

When the data values change, the colors change abruptly, yielding easily detectable band like patterns in the visualization.

Instead of a height plot, we now use a grayscale colormap that maps the function value range [0,1] to 30 distinct shades of gray.

If we use instead a colormap that maps the same value range to an alternating pattern of black and white colors, we obtain the zebra-like result. Thin, dense stripes indicate regions of high variation speed of the function, whereas the flat areas in the center and at the periphery of the image indicate regions of slower variation.

Grayscale colormap. Here, we map data values f linearly to luminance, or gray value, with f_{min} corresponding to black and f_{max} corresponding to white. Most medical specialists, but also nonspecialists, would agree that the grayscale produces a much easier-to-follow, less-confusing visualization on which details are easier to spot than when using the Rainbow Colormap Visualization

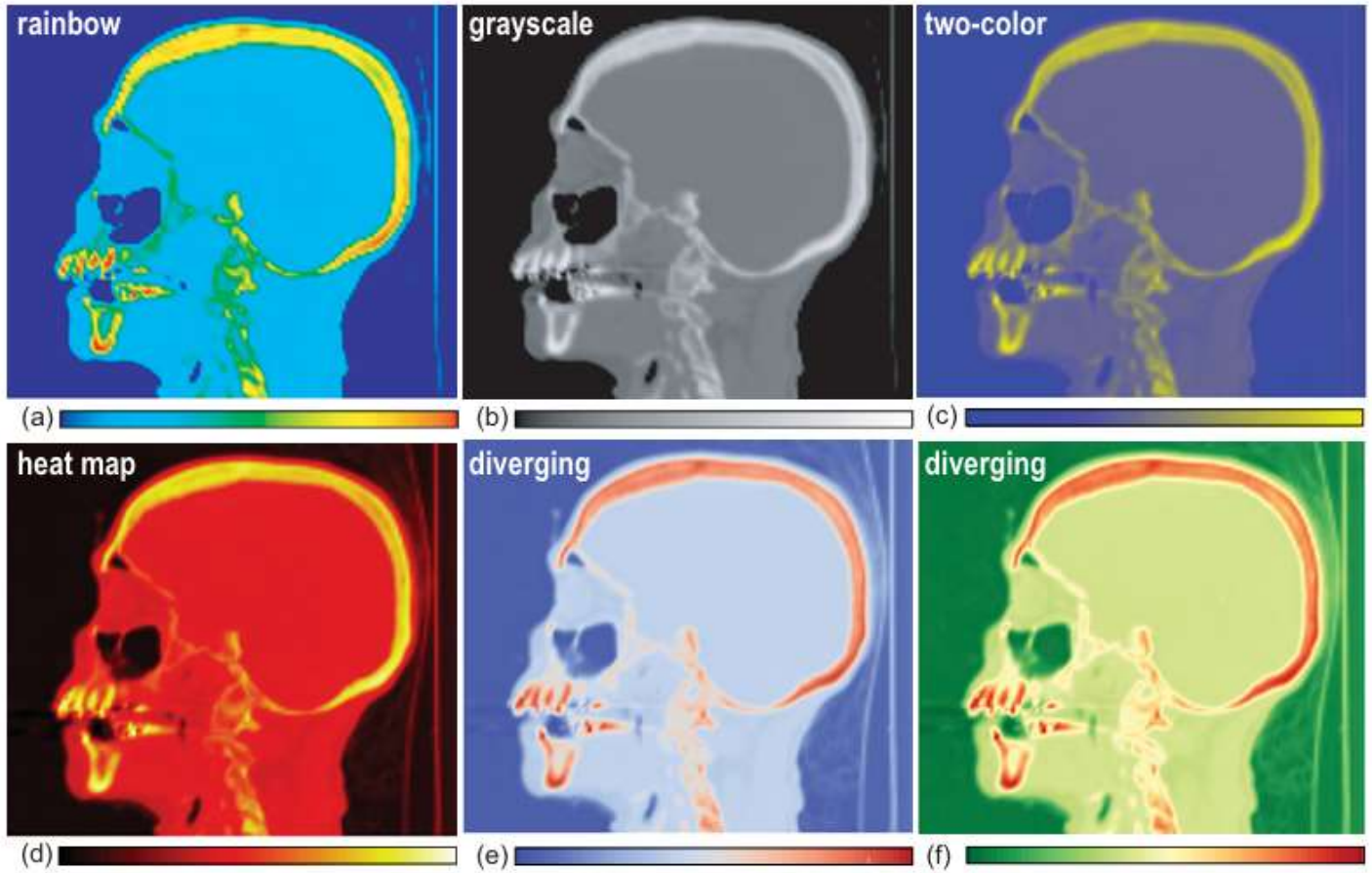


luminance color map rainbow color map

Images courtesy of Alexandru Telea

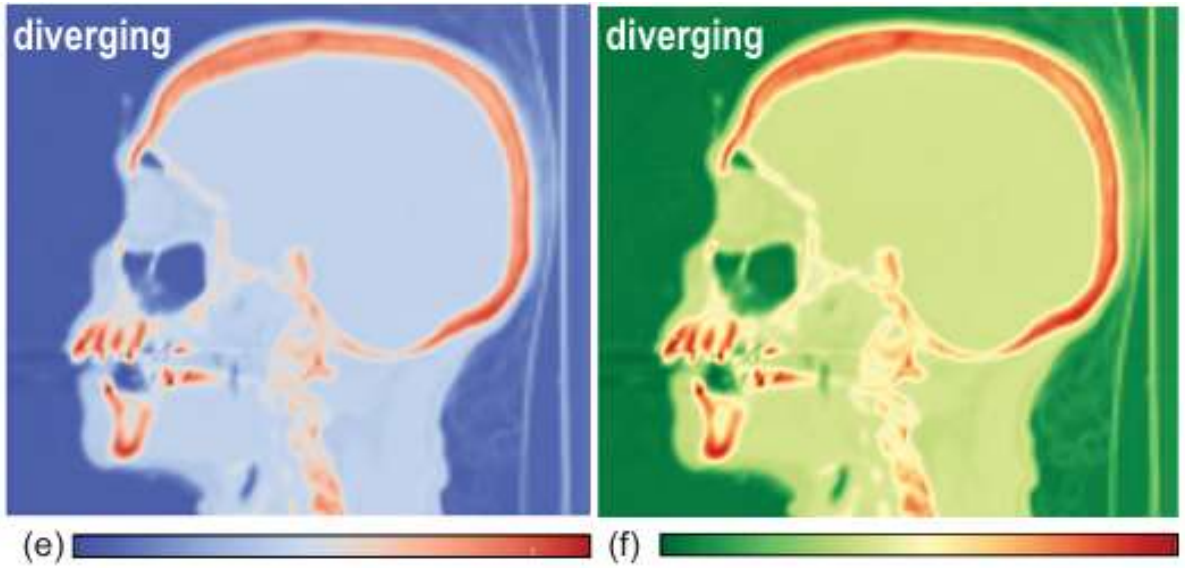
PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
WRIGHT STATE UNIVERSITY
40 YEARS

Scaler Visualizations with various Color Maps



(c) A two-hue colormap. The colormap entries are obtained by linearly interpolating between two user-selected colors, blue and yellow in our case.

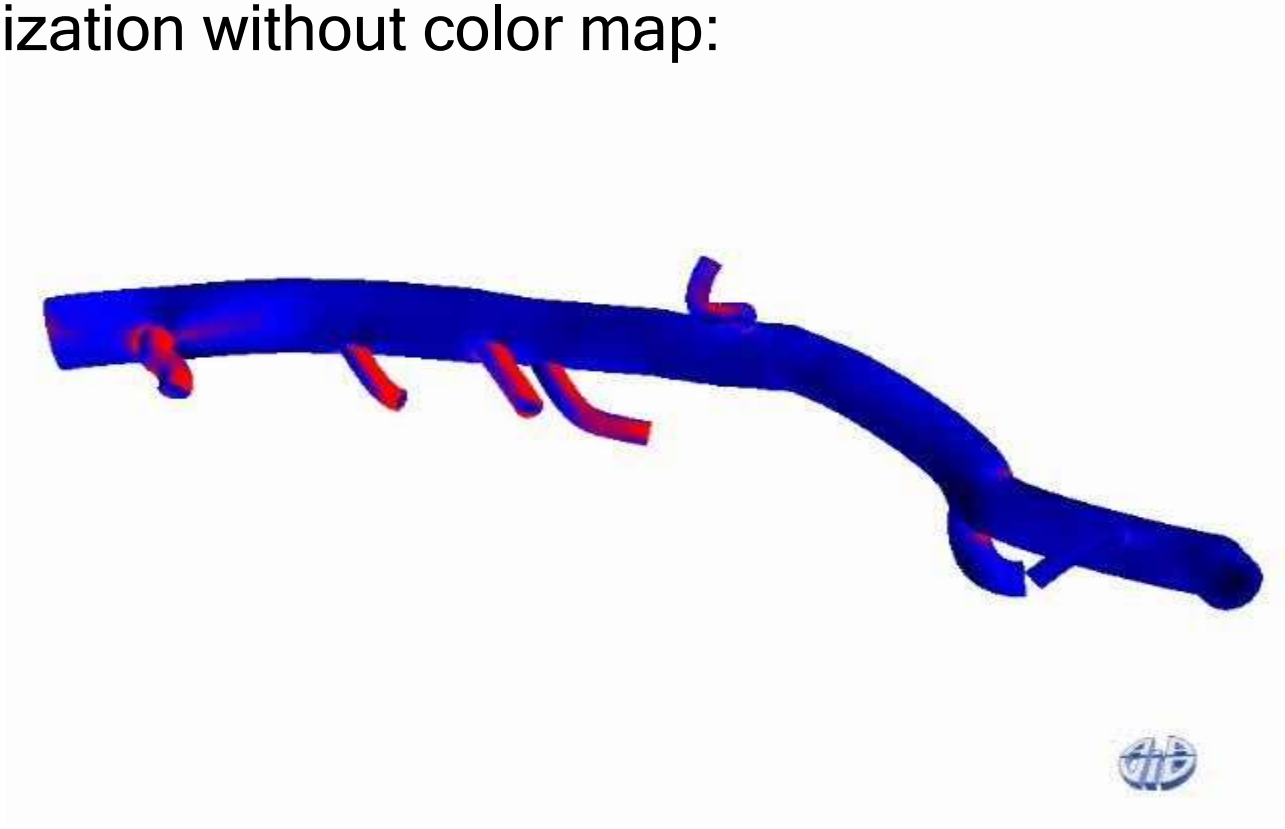
(d) A heat map, or heated body colormap. Black corresponding to low data values, red-orange hues for intermediate data ranges, and yellow-white hues for the high data values respectively.



(e,f) Rather than interpolating between the end colors cmin and cmax, we now add a third color cmid for the data value fmid =(fmin + fmax)/2 located in the middle of the considered data range [fmin,fmax],

Scalar Visualization

Visualization without color map:

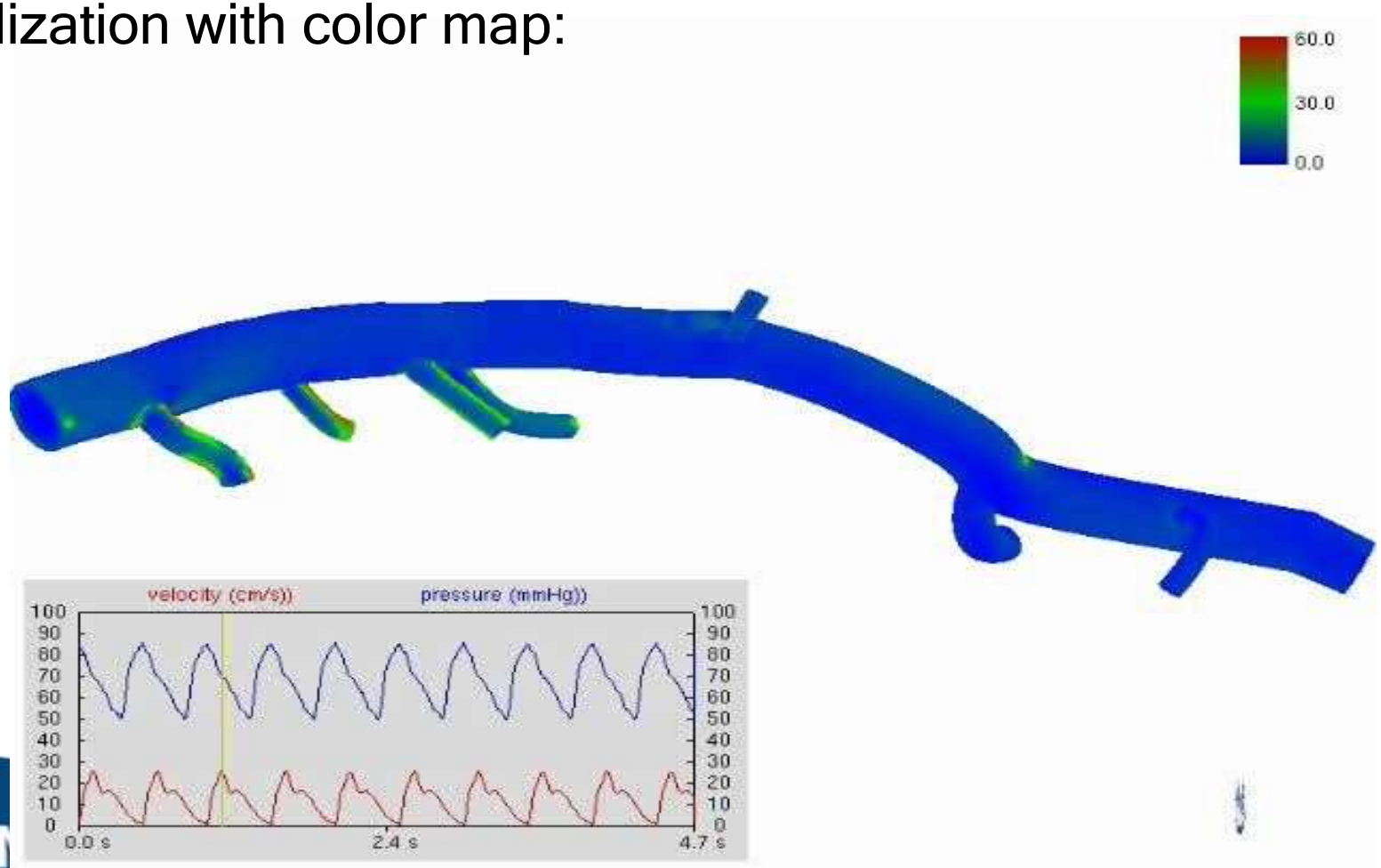


PRESIDENCY UNIVERSITY
GAIN MORE KNOWLEDGE REACH GREATER HEIGHTS
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM

a legend explaining how the colors are mapped to the original values. Without this information, the color information can be almost useless as the frame of reference is missing. Providing this context information can help better interpret the image visualizing the data.

Scalar Visualization

Visualization with color map:



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 11 of 2013
WRIGHT STATE UNIVERSITY
40 YEARS

Scalar Visualization

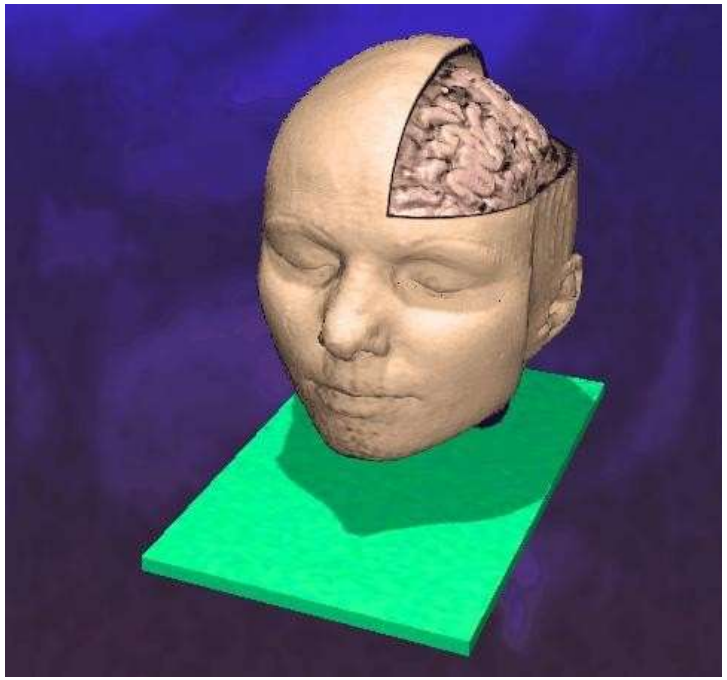
Contouring

- A natural extension to color mapping is contouring.
- When we see a surface colored with data values, the eye often separates similarly colored areas into distinct regions.
- When we contour data, we are effectively constructing the boundary between these regions. These boundaries correspond to contour lines (2-D) or surfaces (3-D) of constant scalar value.
- Examples of 2-D contour displays include weather maps annotated with lines of constant temperature (isotherms), or topological maps drawn with lines of constant elevation.

Scalar Visualization

Three-dimensional contours are called *isosurfaces*, and can be approximated by many polygonal primitives.

Examples of isosurfaces include constant medical image intensity corresponding to body tissues such as skin, bone, or other organs. (The corresponding isovalue for the same tissue, however, is not necessarily constant among several different scans.) Other abstract isosurfaces such as surfaces of constant pressure or temperature in fluid flow also may be created.



Scalar Visualization

Height Plots

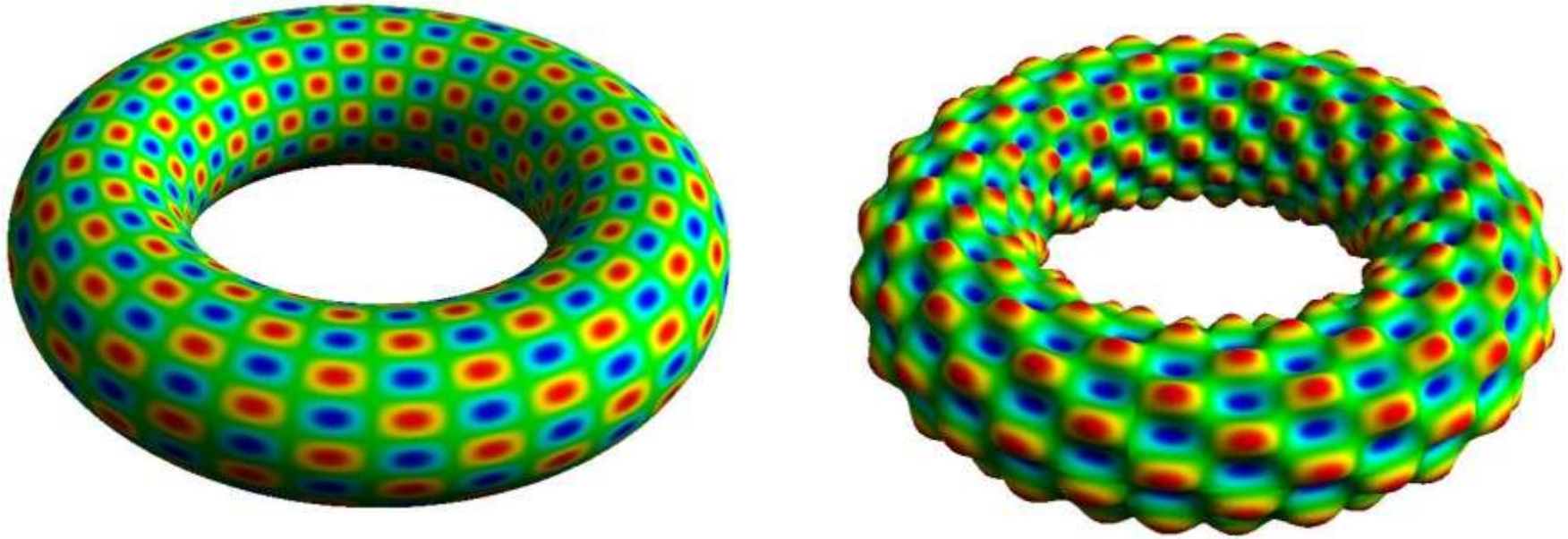
Height plots, also called elevation or carpet plots, were introduced by our first example during the introduction. Given a two-dimensional surface $D_s \in D$, part of a scalar dataset D, height plots cam be described by the mapping operation

$$m: D_s \rightarrow D,\ m(x)=x+s(x)n(x),\ \forall x \in D_s$$

Where $s(x)$ is the scalar value of D at the point x and $n(x)$ is the normal to the surface D_s at x. In other words, the height plot mapping operation "warps" a given surface D_s included in the data set along the surface normal with a factor proportional to the scalar values.

Scalar Visualization

Example: a torus surface (a) and its “warped” variant with the height corresponding to the scalar value.



(a) (b)

Images courtesy of Alexandru Telea

Vector Visualization

Vector data is a three-dimensional representation of direction and magnitude. Vector data often results from the study of fluid flow, or when examining derivatives, i.e. rate of change, of some quantity.

Different visualization techniques are available for vector data sets, for example:

- Hedgehogs and oriented glyphs
- Warping
- Displacement plots
- Time animation
- Streamlines

Vector Glyph

$$l = (\vec{x}, \vec{x} + k\vec{V}(\vec{x}))$$

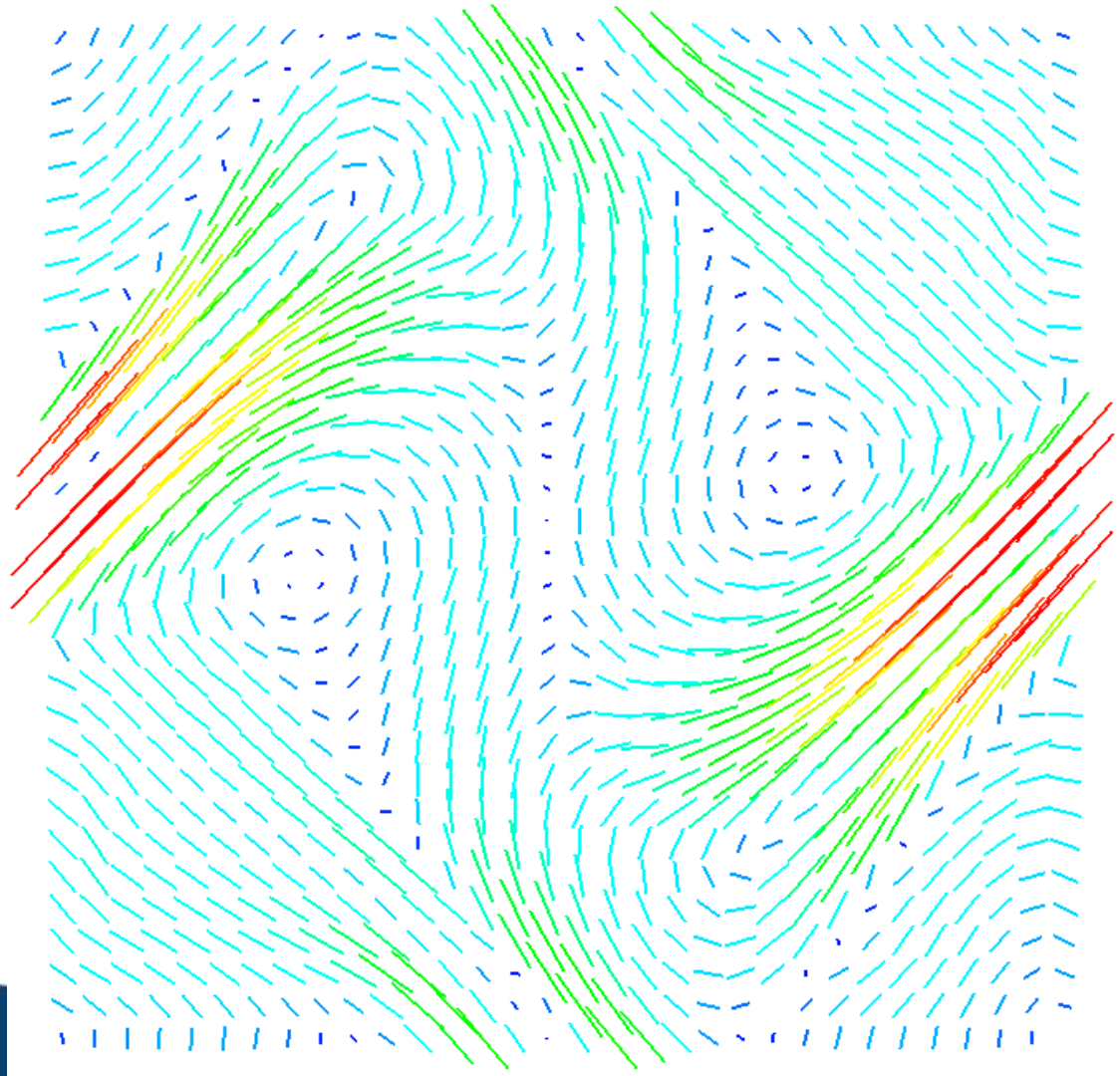
- Vector glyph mapping technique associates a vector glyph (or icon) with the sampling points of the vector dataset
- The magnitude and direction of the vector attribute is indicated by the various properties of the glyph: location, direction, orientation, size and color.
- Many variations of framework
  - Lines (convey direction)
  - 3D cone (convey direction + orientation)
  - Arrow (convey direction + orientation)

Vector Glyph

Line glyph,
or hedgehog glyph

Sub-sampled by
a factor of 8
(32 X 32)

Original (256 X 256)



Velocity Field of a 2D Magnetohydrodynamic Simulation

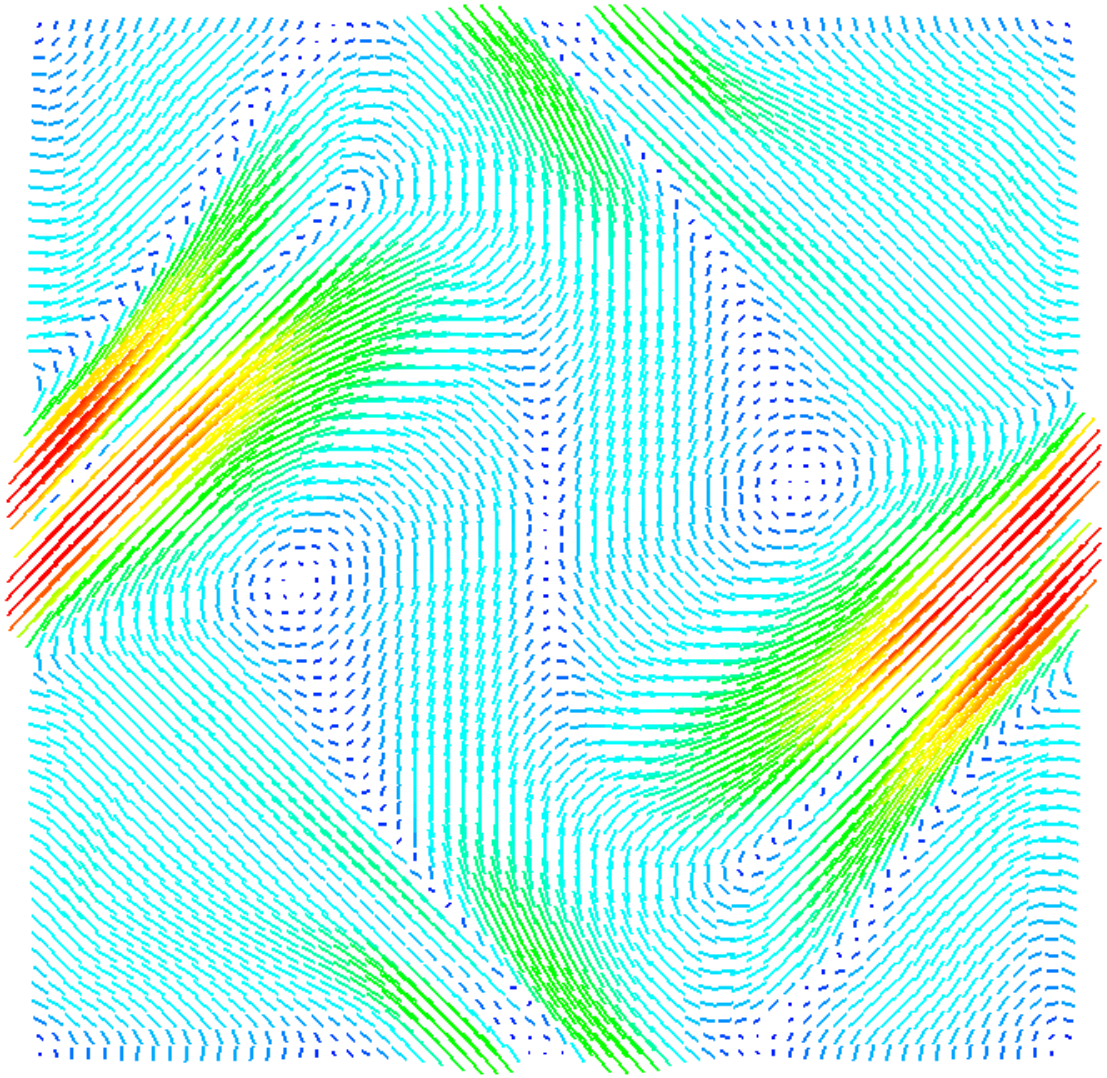
PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka St

Vector Glyph

Line glyph,
or hedgehog glyph

Sub-sampled by
a factor of 4
(64 X 64)

Original (256 X 256)



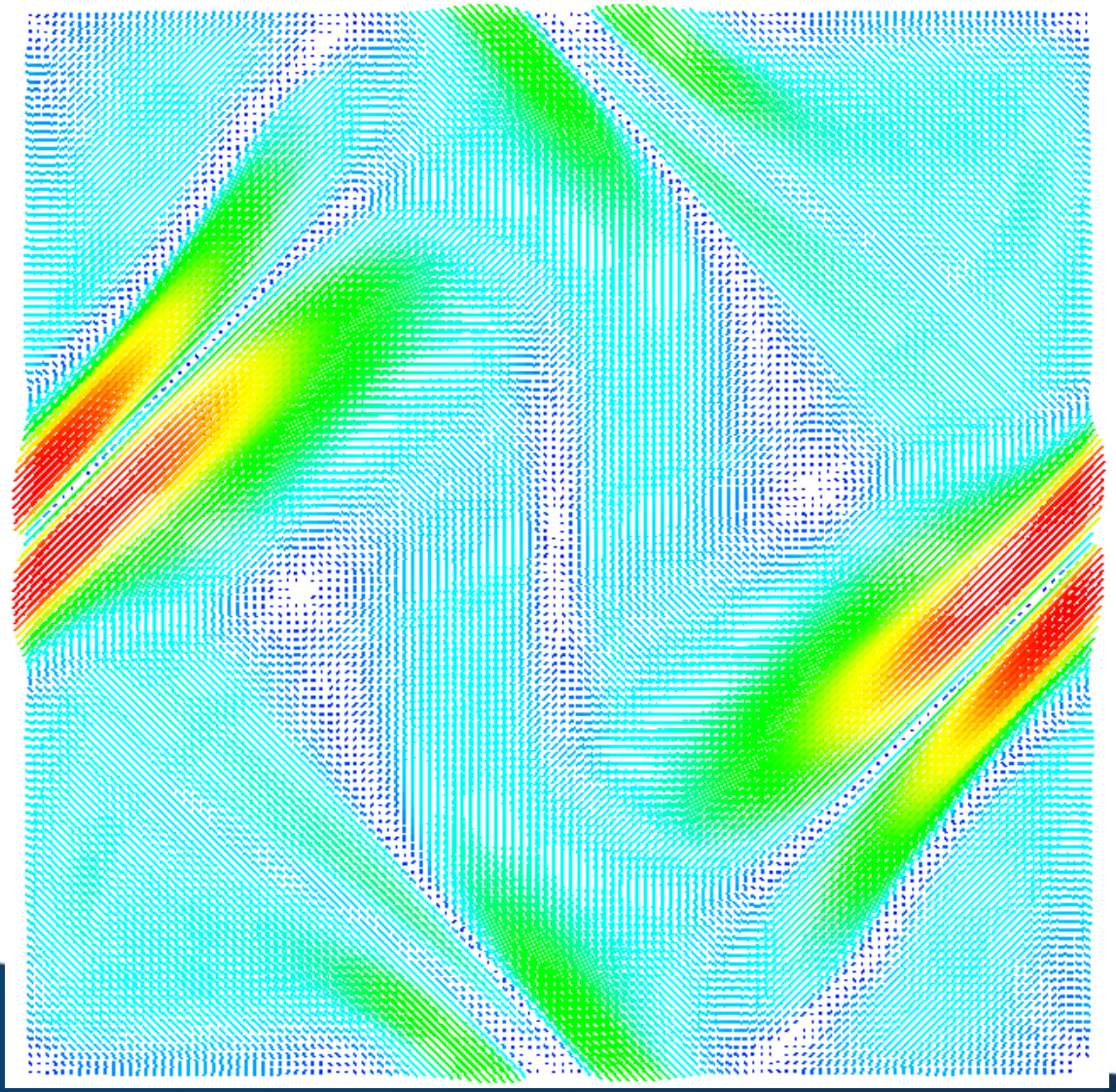
PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS

Vector Glyph

Sub-sampled by a factor of 2 (128 X 128)

Original (256 X 256)

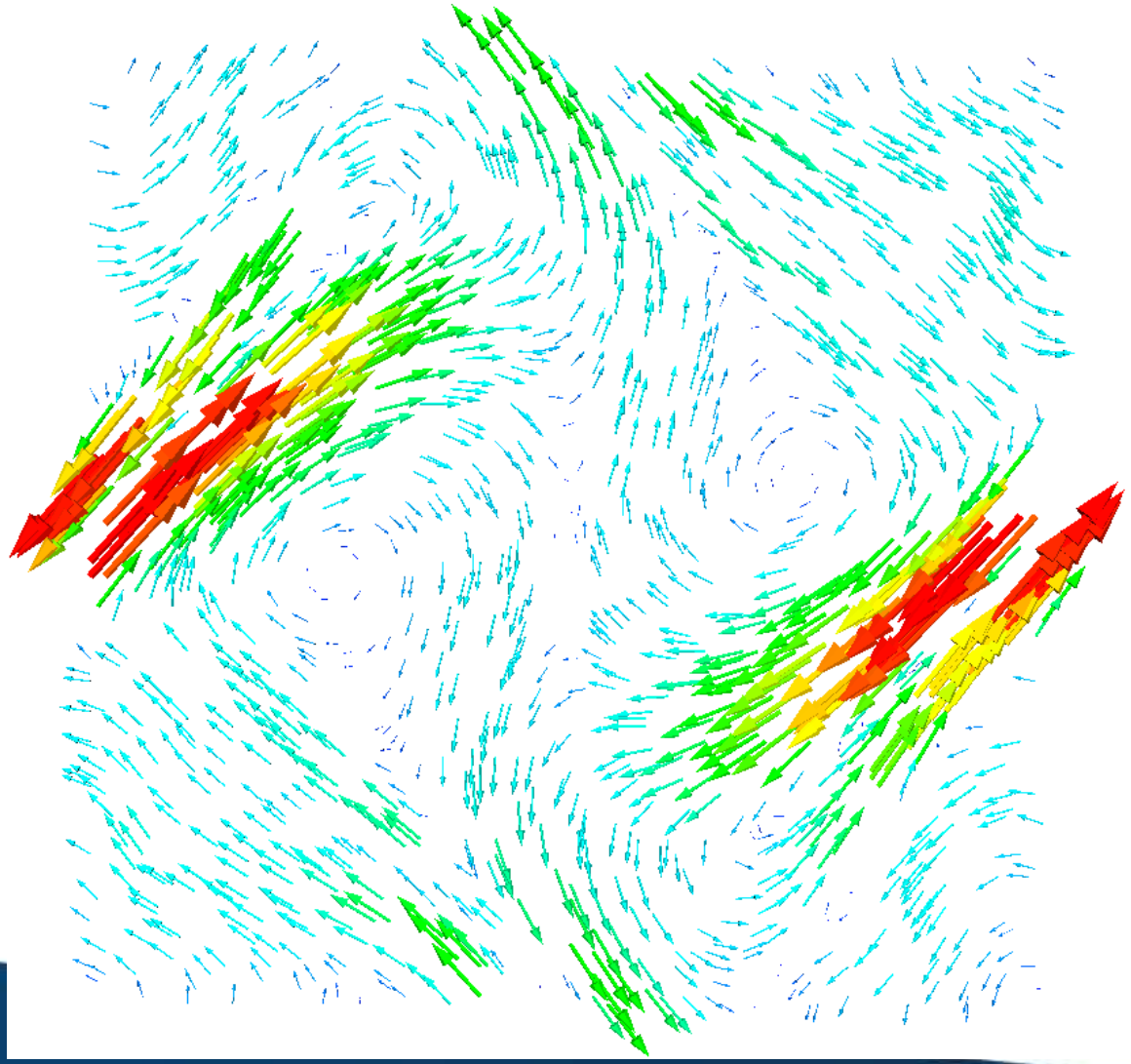
Problem with a dense representation using glyph:
(1) clutter
(2) miss-representation



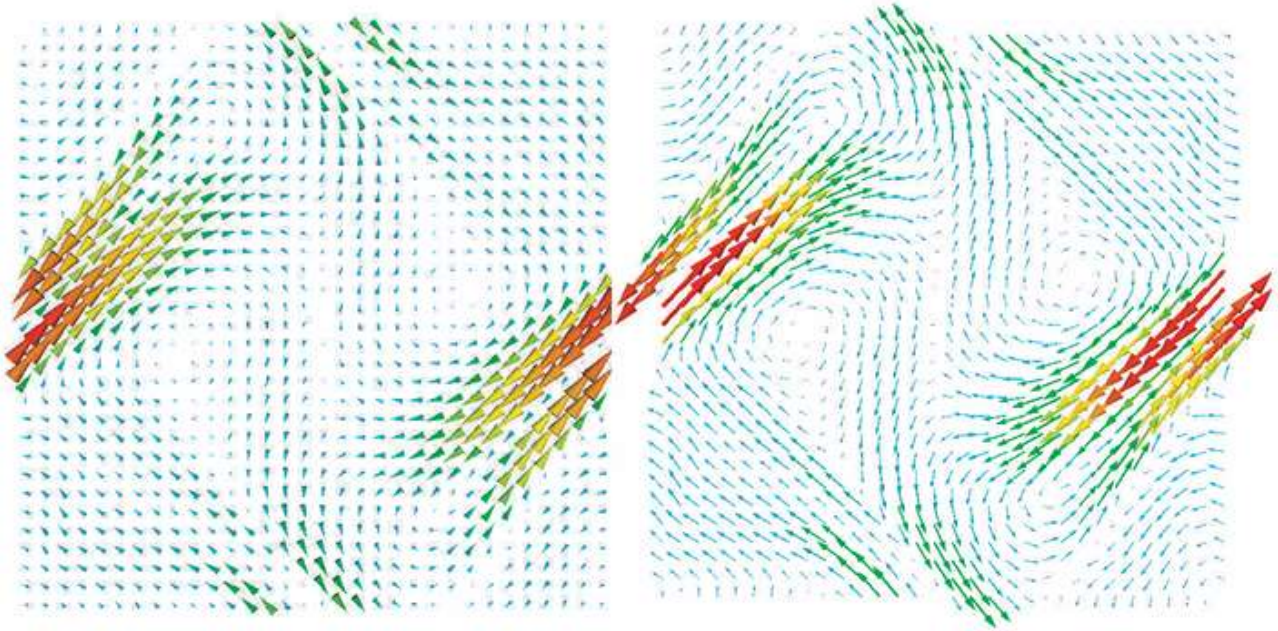
PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS

Vector Glyph

Random
Sub-sampling
Is better



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS



(a) (b)

Figure 6.5. Different glyph types. (a) Cones. (b) Arrows.

Vector Glyph

- Glyph method is simple to implement, and intuitive to interpretation
- High-resolution vector datasets must be sub-sampled in order to avoid overlapping of neighboring glyphs.
- Glyph method is a sparse visualization: does not represent all points
- Occlusion
- Subsampling artifacts: difficult to interpolate
- Alternative: color mapping method is a dense visualization

Vector Color Coding

- Similar to scalar color mapping, vector color coding is to associate a color with every point in the data domain
- Typically, use HSV(Hue-Saturation-Value) system in which a color wheel is used
  - Hue is used to encode the direction of the vector, e.g., angle arrangement in the color wheel
  - Value of the color vector is used to encode the magnitude of the vector
  - Saturation is set to one

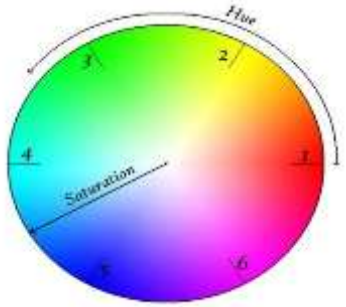
For example, a bright red vector may indicate a strong wind blowing to the east, while a dimmer red vector might represent a weaker eastward wind.

This technique is particularly useful for visualizing vector fields in a way that makes it easy to discern both the direction and relative strength of vectors at different points in the data. It enhances the interpretability of complex vector data and is commonly used in various scientific and engineering disciplines.

Vector Color Coding

2-D Velocity Field of the MHD simulation:

Orientation, Magnitude

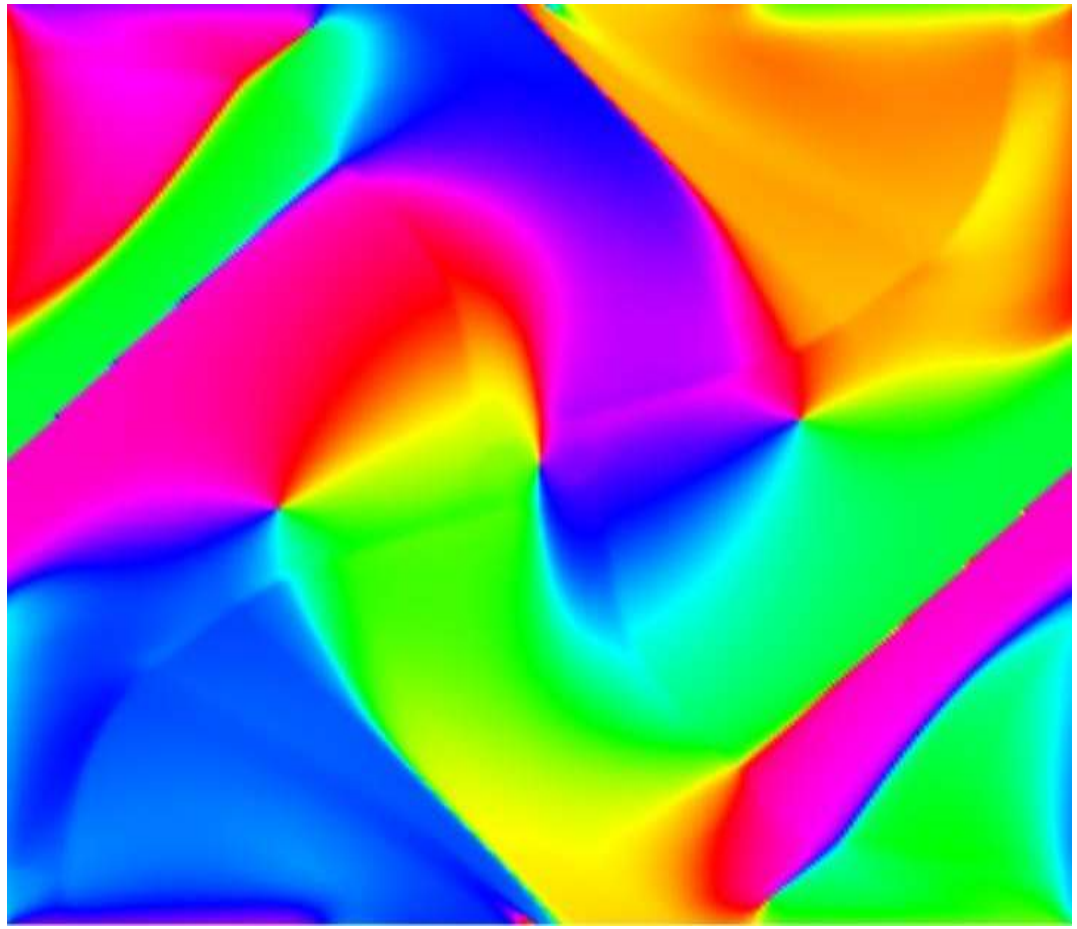


Every distinct hue corresponds to a different angle of the color wheel: red is 0◦, magenta is 60◦, blue is 120◦, cyan is 180◦, green is 240◦, and yellow is 300◦.

Vector Color Coding

2-D Velocity Field of the MHD simulation:

Orientation only; no magnitude



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS

Visualization Techniques on Trees, Graphs & Networks

Introduction

Important application of visualization is the conveying of relational information, e.g., how data items or records are related to each other. These interrelationships can take many forms:

- part/subpart, parent/child, or other hierarchical relation;
- connectedness, such as cities connected by roads or computers connected by networks;
- derived from, as in a sequence of steps or stages;
- shared classification;
- similarities in values;
- similarities in attributes (e.g., spatial, temporal).

Introduction Cont…

- Relationships can be simple or complex:
- unidirectional or bi-directional
- nonweighted or weighted
- certain or uncertain

Indeed, the relationships may provide more and richer information than that contained in the data records.

Applications for visualizing relational information are equally diverse, from categorizing biological species, to exploring document archives, to studying a terrorist network.

Displaying Hierarchical Structures (Trees)

Trees or hierarchies (we'll use the terms interchangeably) are one of the most common structures to hold relational information. We can divide these techniques into two classes of algorithms:

- space-filling
- non-space-filling

PRESIDENCY UNIVERSITY
GAIN MORE KNOWLEDGE REACH GREATER HEIGHTS
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Space Filling Method

- Space-filling techniques make maximal use of the display space. This is accomplished by using juxtapositioning to imply relations, as opposed to, for example, conveying relations with edges joining data objects.
- The two most common approaches to generating space-filling hierarchies are rectangular and radial layouts.

1. ***Treemap***, a rectangle is recursively divided into slices, alternating horizontal and vertical slicing, based on the populations of the subtrees at a given level.

2. ***Sunburst displays***, have the root of the hierarchy in the center of the display and use nested rings to convey the layers of the hierarchy. Each ring is divided based on the number of nodes at that level.

Space Filling Method

Pseudo code for treemap:

Start : Main Program

  Width = width of rectangle

  Height = height of rectangle

  Node = root node of the tree

  Origin = position of rectangle, e.g. , **[0,** 0]

  Orientation = direction of cuts, alternating between horizontal and vertical

  Treemap(Node, Orientation, Origin, Width, Height)

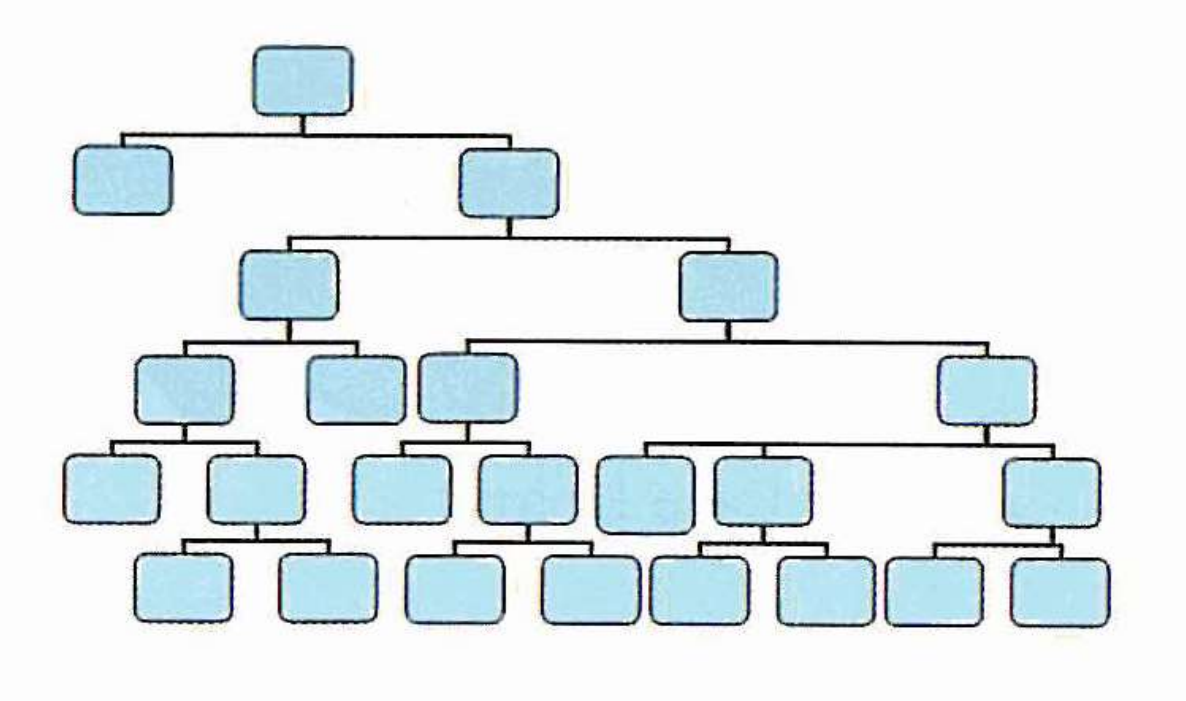
End: Main Program

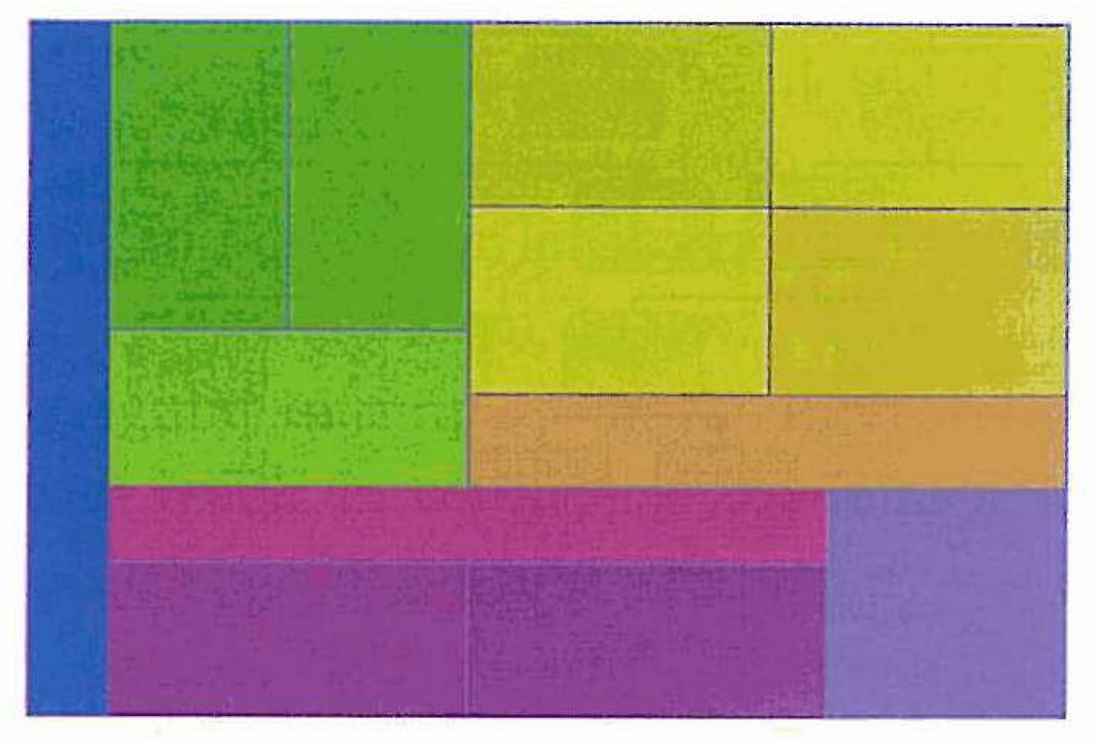
Space Filling Method

```
Treemap(node n, orientation o, position orig, hsize W, vsize h)
 if n is a terminal node (i.e., it has no children)
    draw-rectangle( orig, W, h)
     return
for each child of n (child-i), get number of terminal nodes in subtree
sum up number of terminal nodes
compute percentage of terminal nodes in n from each subtree (percent-i)
if orientation is horizontal
    for each subtree
        compute offset of origin based on origin and width (offset-i)
        treemap(chi1d-i, vertical, orig + offset-i, W 9 percent-i, h)
else
    for each subtree
        compute offset of origin based on origin and height (offset-i)
        treemap(chi1d-i, horizontal, orig + off set-i, W, h * percent-i)
End: Treemap
```

PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

Space Filling Method





PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

Space Filling Method

Pseudo code for sunburst display

Start : Main Program
Start = start angle for a node (initially 0)
End = end angle for a node (initially 360)
Origin = position of center of sunburst, e.g., [0,0]
Level = current level of hierarchy (initially 0)
Width = thickness of each radial band - based on max depth and display size
Sunburst(Node, Start, End, Level)
End: Main Program

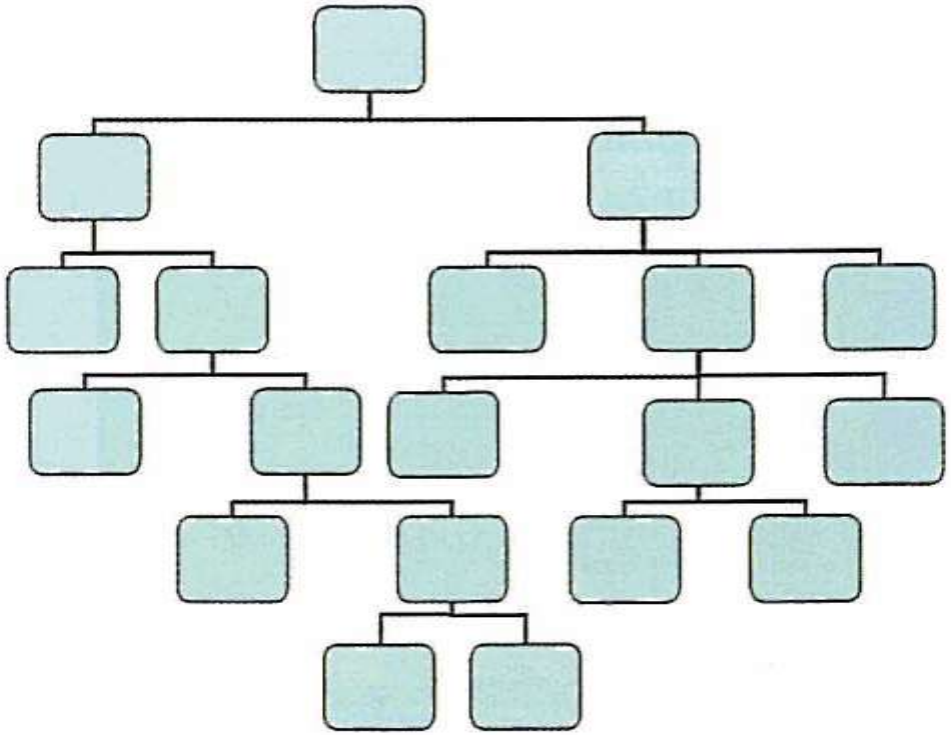
PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

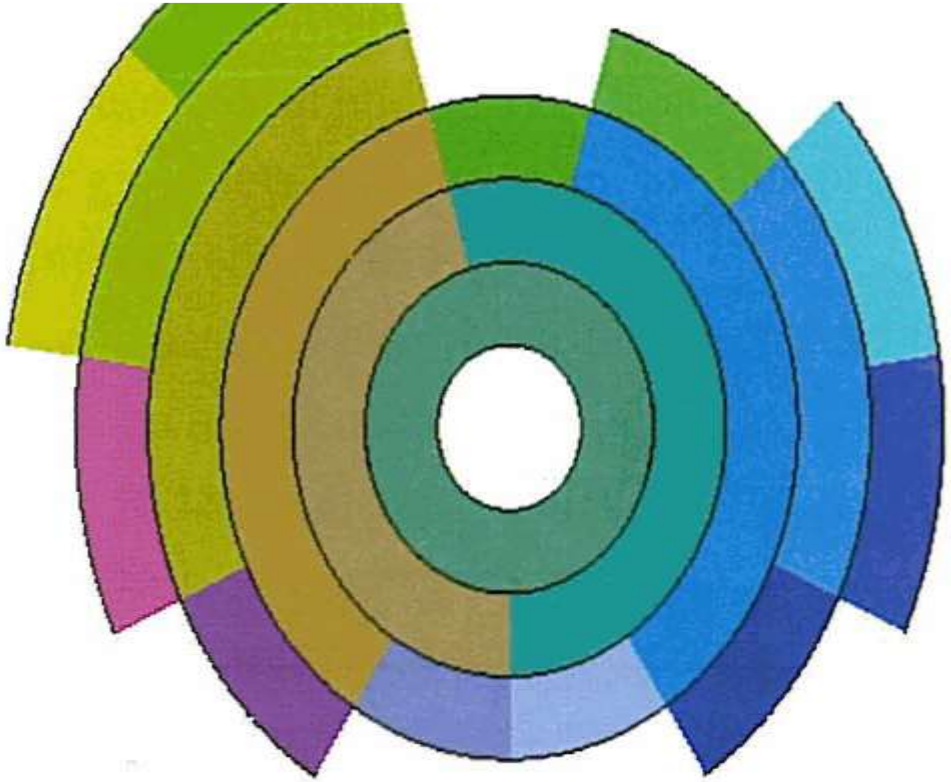
Space Filling Method

Pseudo code for sunburst display

Sunburst (node n, angle st, angle en, level 1)
if n is a terminal node (i.e., it has no children)
  draw-radial-section(Origin, st, en, l * Width, (1+i) * Width)
  return
for each child of n (child-i), get number of terminal nodes in subtree
sum up number of terminal nodes
compute percentage of terminal nodes in n from each subtree (percent31
for each subtree
   compute start/end angle based on size of subtrees, order, and angle range
   Sunburst (child-i, st-i, en-i, 1+i)
End: Sunburst

Space Filling Method





PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Non-Space Filling Methods

The most common representation used to visualize tree or hierarchical relationships is a *node-link diagram.*

The drawing of such trees is influenced the most by two factors:

- The fan-out degree (e.g., the number of siblings a parent node can have)
- The depth (e.g., the furthest node from the root)

➢ When designing an algorithm for drawing any node-link diagram (not just trees), one must consider three categories of often-contradictory guidelines:
  ➢ drawing conventions,
  ➢ constraints, and
  ➢ aesthetics.

➢ Conventions may include restricting edges to be either a single straight line, a series of rectilinear lines, polygonal lines, or curves.

➢ Constraints may include requiring a particular node to be at the center of the display, or that a group of nodes be located close to each other, or that certain links must either go from top to bottom or left to right.

Each of the above guidelines can be used to drive the algorithm design.

Non-Space Filling Methods

Aesthetics, however, often have significant impact on the interpretability of a tree or graph drawing, yet often result in conflicting guidelines. Some typical aesthetic rules include:

- minimize line crossings
- maintain a pleasing aspect ratio
- minimize the total area of the drawing
- minimize the total length of the edges
- minimize the number of bends in the edges
- minimize the number of distinct angles or curvatures used
- strive for a symmetric structure

Non-Space Filling Methods

A simple tree drawing procedure is given below:

- Slice the drawing area into equal-height slabs, based on the depth of the tree.
- For each level of the tree, determine how many nodes need to be drawn.
- Divide each slice into equal-sized rectangles based on the number of nodes at that level.
- Draw each node in the center of its corresponding rectangle.
- Draw a link between the center-bottom of each node to the center-top
- of its child node(s).

Displaying Arbitrary Graphs/Networks

- A Graph is a non-linear data structure consisting of vertices and edges.
- The vertices are sometimes also referred to as nodes and the edges are lines or arcs that connect any two nodes in the graph.
- More formally a Graph is composed of a set of vertices(V) and a set of edges(E). The graph is denoted by G(E, V).

Components of a Graph

- **Vertices:** Vertices are the fundamental units of the graph. Sometimes, vertices are also known as vertex or nodes. Every node/vertex can be labeled or unlabelled.
- **Edges:** Edges are drawn or used to connect two nodes of the graph. It can be ordered pair of nodes in a directed graph. Edges can connect any two nodes in any possible way.

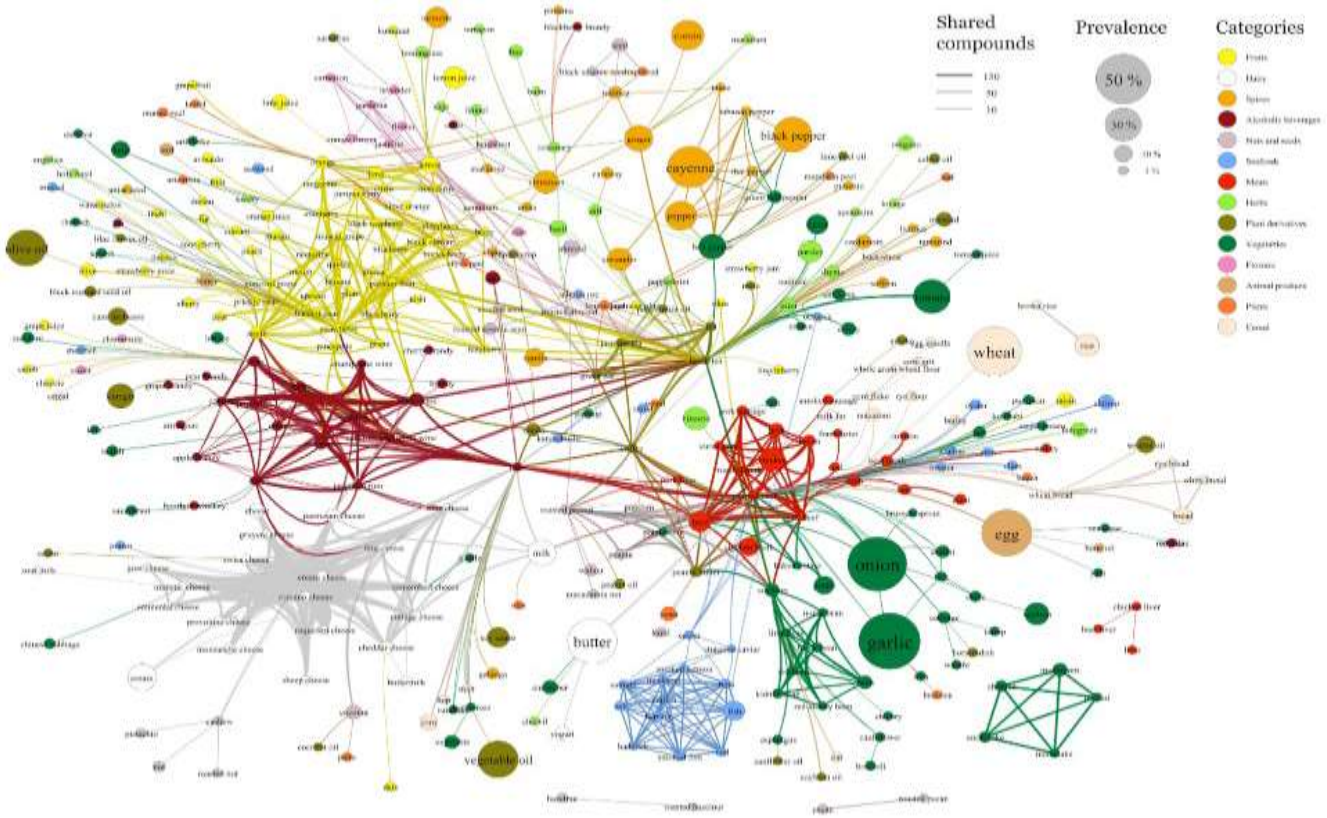
Displaying Arbitrary Graphs/Networks

Node-Link **Graphs:**

- A cutvertex is any node that causes the graph to be disconnected if it is removed.
- A biconnected graph is one without a cutvertex.
- A block is a maximally biconnected subgraph of a graph.
- A separating pair means two vertices whose removal causes a biconnected graph to become disconnected.
- A triconnected graph is one without a separating pair. A planar triconnected graph has a unique embedding.

Displaying Arbitrary Graphs/Networks

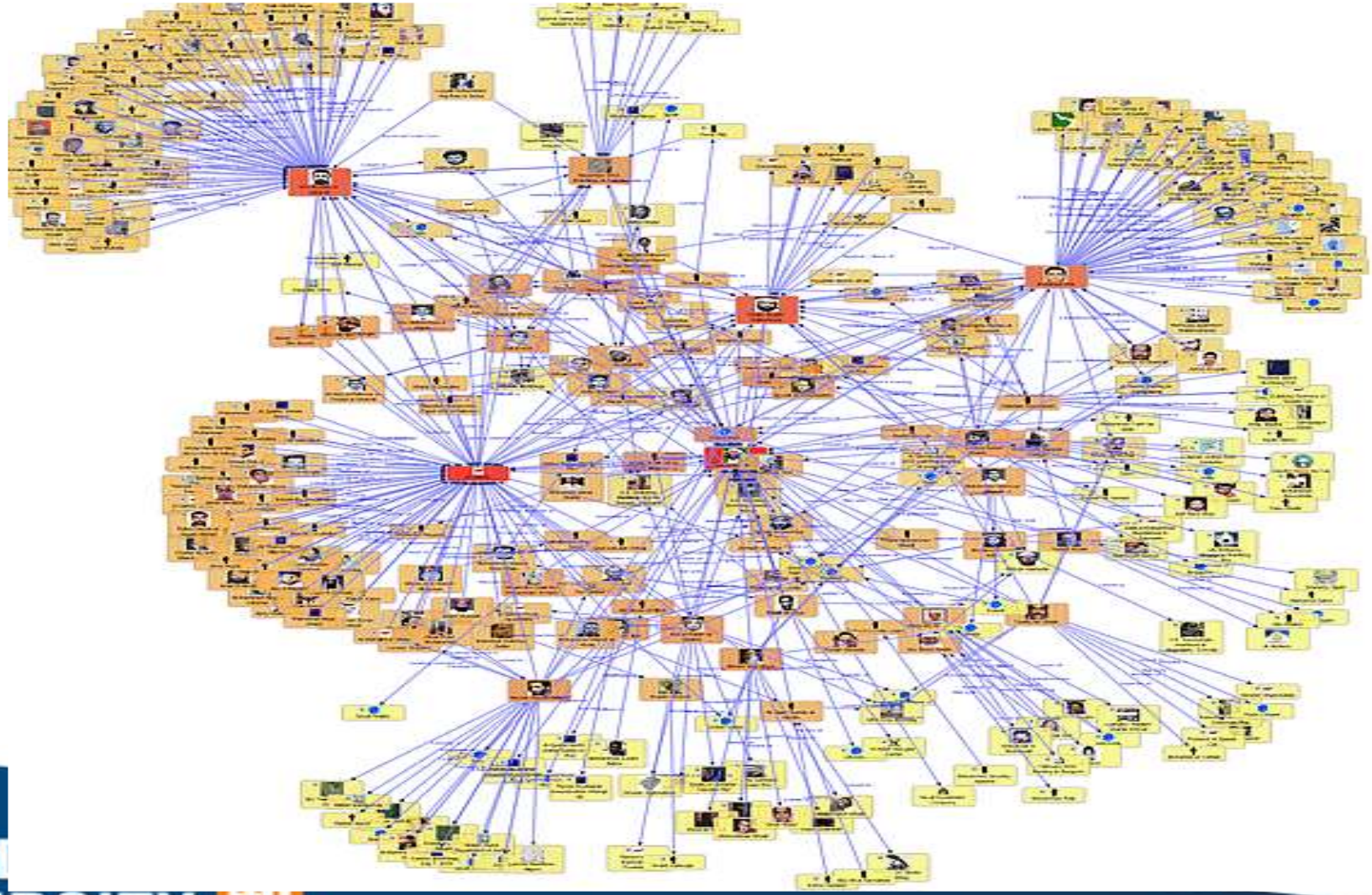
Node-Link **Graphs:**



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Displaying Arbitrary Graphs/Networks

Node-Link **Graphs:(Social Network)**



PRESID
UNIVERSITY
Private University Estd. in Karnataka State by Act No. 11 of 2013
WRIGHT STATE UNIVERSITY
40 YEARS OF ACADEMIC WISDOM

Displaying Arbitrary Graphs/Networks

Matrix Representations for Graphs:

- An alternate visual representation of a graph is via an adjacency *matrix*, which is an N by N grid (where N is the number of nodes), where position (i, j) represents the existence (or not) of a link between nodes ***i*** and j.
- This may be a binary matrix, or the value might represent the strength or weight of the link between the two nodes.
- This method overcomes one of the biggest problems with node-link diagrams, namely that of crossing edges, though it doesn't scale well to graphs with large numbers (thousands) of nodes.

Displaying Arbitrary Graphs/Networks

Matrix Representations for Graphs:

	a	b	c	d	e	f	g	h
a		●	●			●		
b	●			●		●		
c	●				●		●	●
d	●	●				●		
e			●				●	●
f	●	●		●				
g			●		●			●
h			●		●		●	

Visual Variables

Introduction

- Visual variables are distinctions that we can use to create and differentiate symbols on a map.
- Visual variables are the attributes or properties of a graphical element that can be visually perceived and used to convey information. They are the building blocks of visual design and play a crucial role in visual communication.
- There are 10 visual distinctions available for symbolization: location, size, shape, orientation, focus, arrangement, texture, saturation, hue, and value.
- These visual techniques can be used to create a pleasing aesthetic, convey precise geographic information, and create a visual hierarchy that can be understood by the viewer of the map.

Types: Visual Variable Location

- The location visual variable is the position of the object and the environment.
- Location can be determined in absolute, relative, or cognitive terms.
- In any case, location determines where in our environment the object exists.
- No matter whether the data is qualitative or quantitative in nature in order to be map it must have a location.

Types: Qualitative

- Qualitative visual variables are used for nominal data.
- The goal of qualitative visual variables is to show how entities differ from each other.
- Qualitative visual variables show the grouping of similar entities.

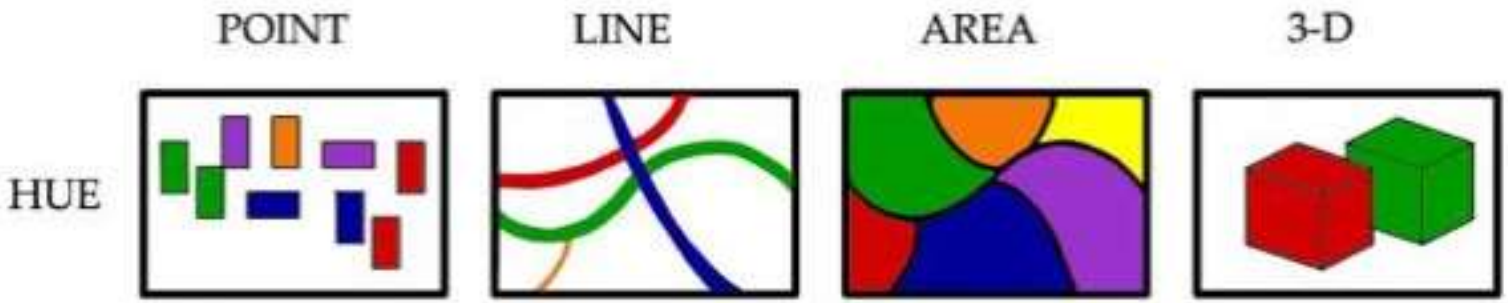
Types: Hue

Hue, more commonly known as color, represents a wavelength on the visible portion of the electromagnetic spectrum.

Hue is great for identifying items as unique, or of a type of item.

Hue creates a perception of groups or likeness.

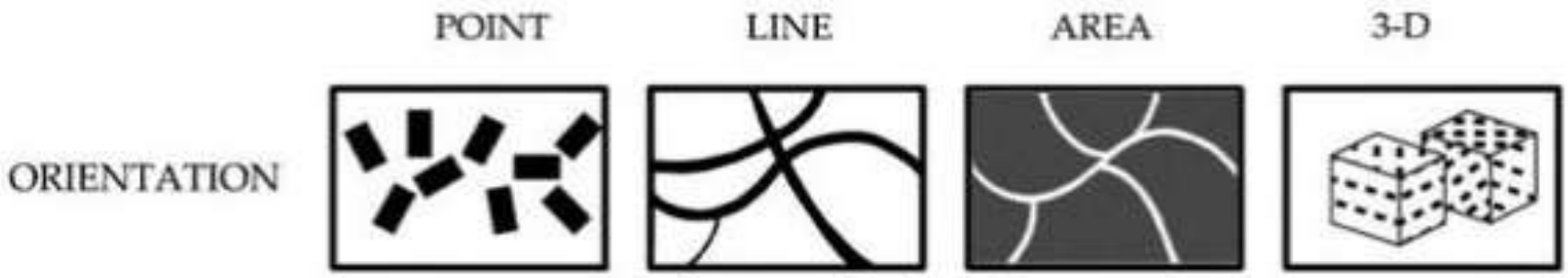
The images show how hue can be applied to data with 0 to 3 dimensions.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

Types: Orientation

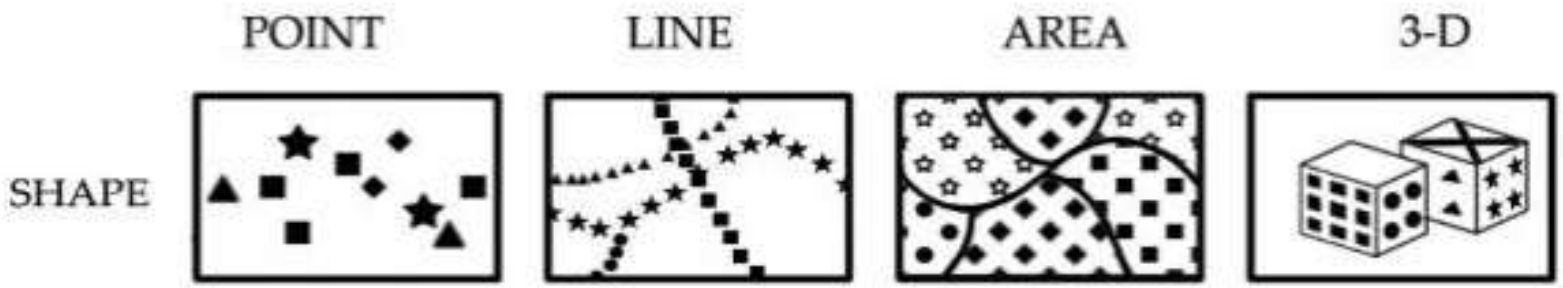
The orientation visual variable changes the orientation of the object and creates a perception of group or likeness.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

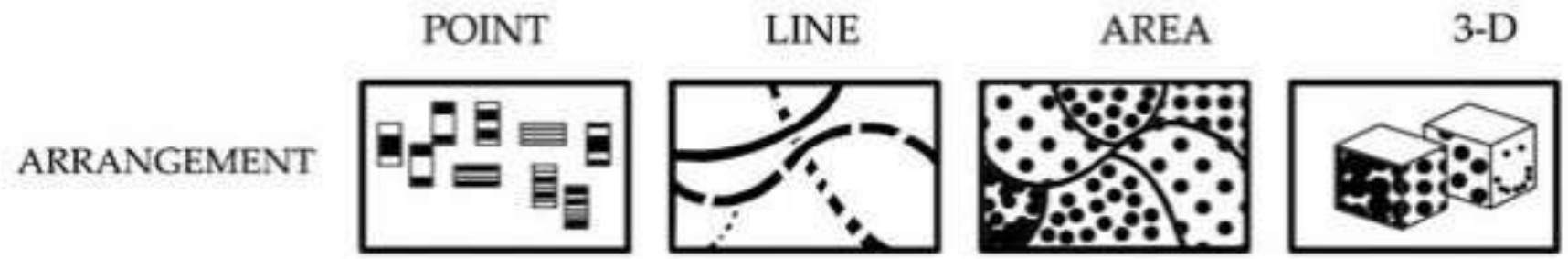
Types: Shape

- The shape visual variable identifies an item as unique or of a type.
- The shape visual variable refers to a point symbol although it can be arranged to resemble a line and placed inside an area or three-dimensional shape.
- The shape does not have to be a geometric form; it can also be a pictorial form.



Types: Arrangement

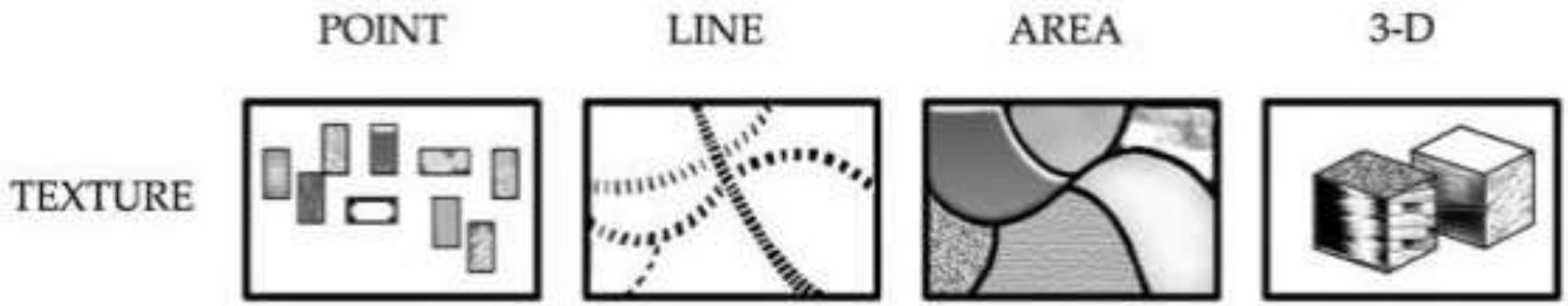
- The arrangement visual variable refers to the placement of elements composing a pattern or a texture.
- Arranging patterns or textures differently can create a perception of a single unique item or a group of items.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

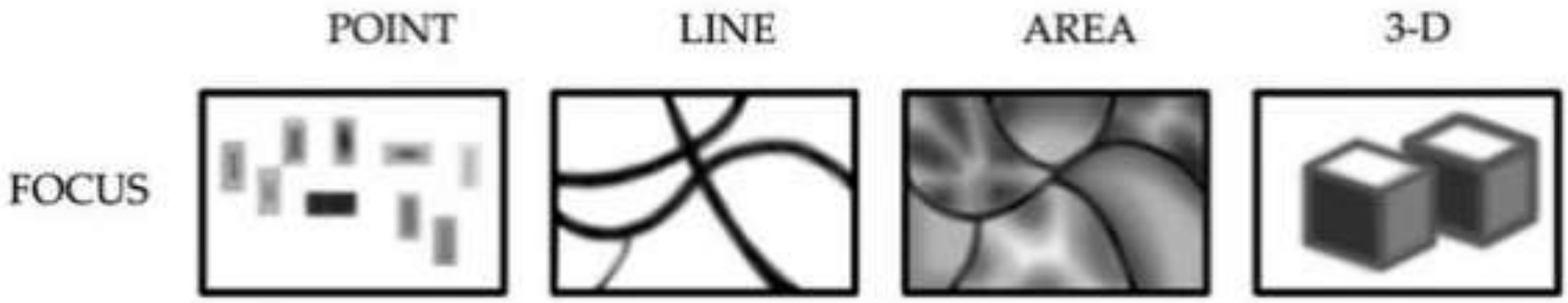
Types: Texture

- Texture refers to the symbols covering an area. Textures identify items as unique or of a type.



Types: Focus

- Focus represents uncertainty by making the symbols look fuzzy or out of focus. The more uncertain a value is the fuzzier or out of focus it should look.

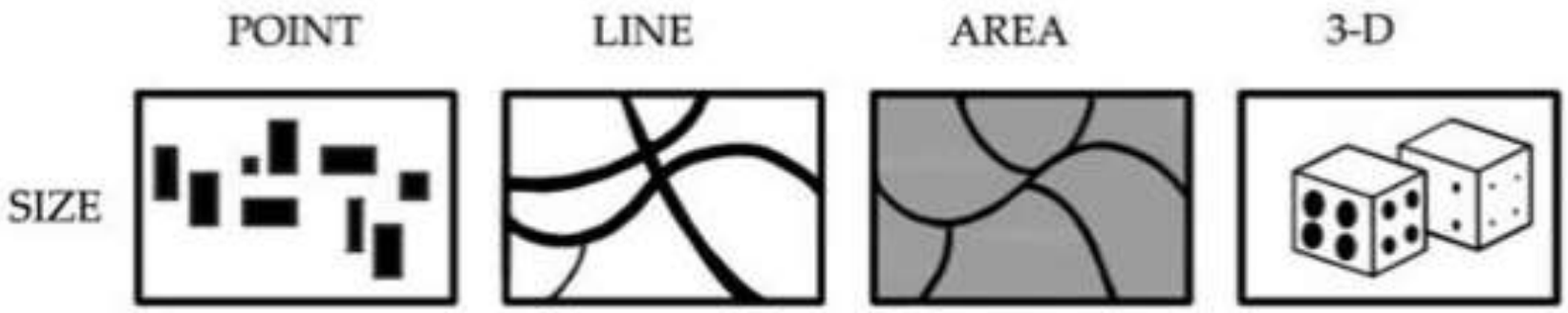


Types: Quantitative Visual Variables

- Quantitative visual variables are used to display ordinal, interval, or ratio data.
- The goal of the quantitative visual variable is to show relative magnitude or order between entities.

Types: Size

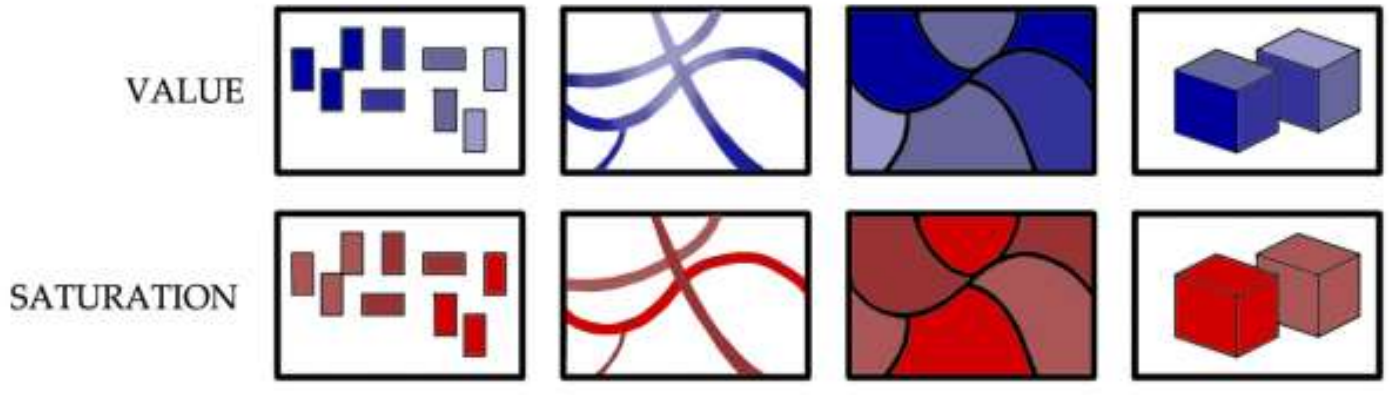
- The size of visual variable changes the size of the symbol to imply relative levels of importance.
- Line thickness implies relative flow levels in the case of road traffic or water flow through a river.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Types: Value, Saturation

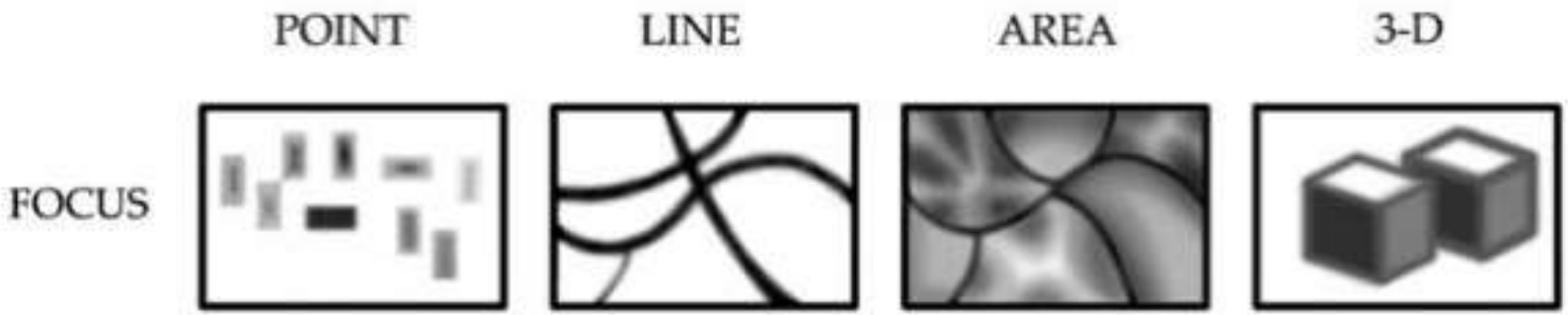
- The visual variable's value and saturation represent different magnitudes or order in a data value. It is important that you only very the saturation or value but not both for a given hue.
- This represents a single variable, which is represented by a single hue, with different quantitative values, which are represented by a difference in saturation or value.

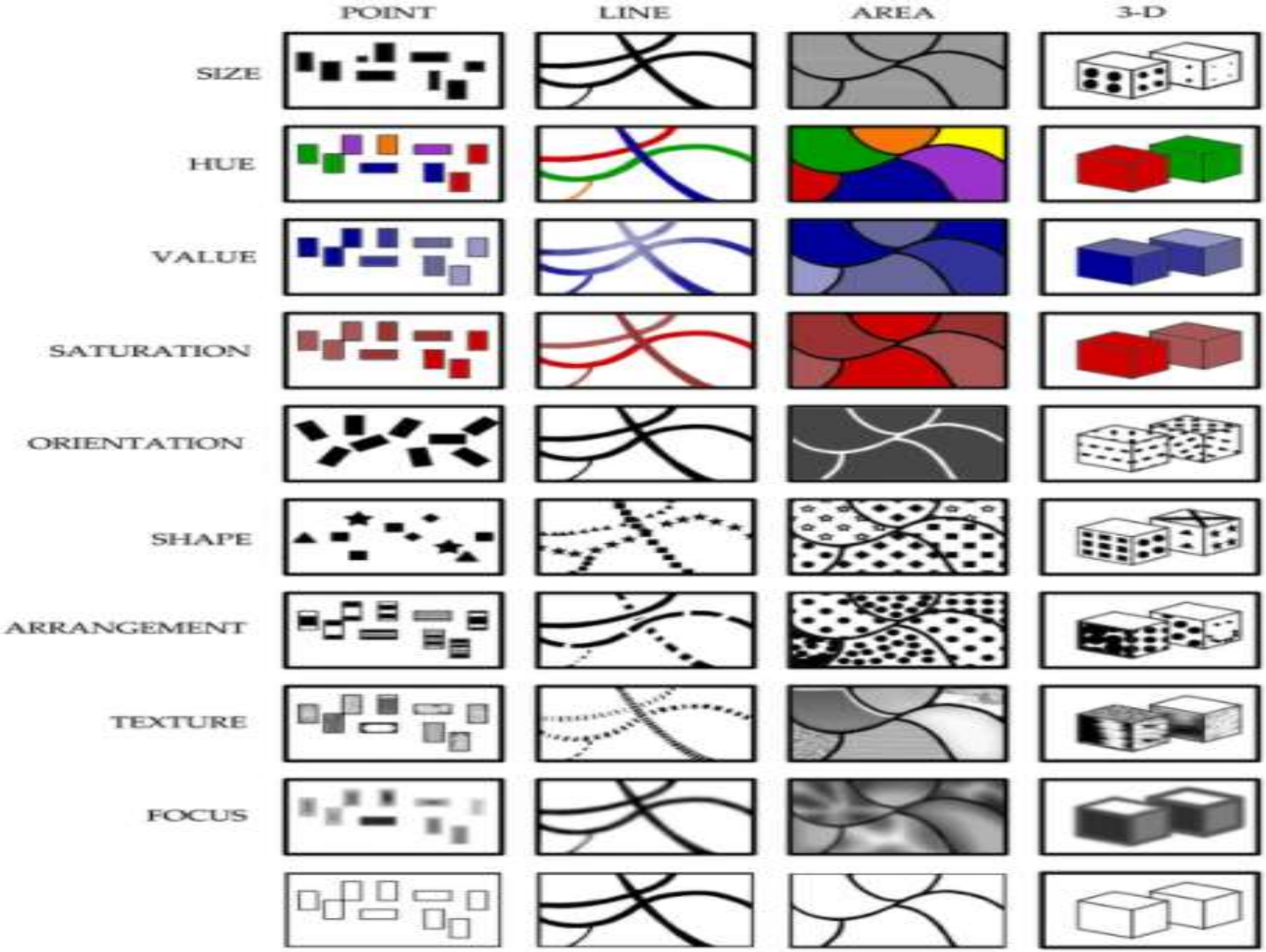


PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

Types: Focus

- Again, the focus visual variable represents uncertainty in quantitative values.




POINT
LINE
AREA
3-D
SIZE
HUE
VALUE
SATURATION
ORIENTATION
SHAPE
ARRANGEMENT
TEXTURE
FOCUS

Map Color & Other Channels

Introduction

<u>Color</u>

- The color is best understood in terms of three separate channels: luminance, hue, and saturation. The major design choice for colormap construction is whether the intent is to distinguish between categorical attributes or to encode ordered attributes.
- Sequential ordered colormaps show a progression of an attribute from a minimum to a maximum value, while diverging ordered colormaps have a visual indication of a zero point in the center where the attribute values diverge to negative on one side and positive on the other

Color

- Bivariate colormaps are designed to show two attributes simultaneously using carefully designed combinations of luminance, hue, and saturation.
- The characteristics of several more channels are also covered: the magnitude channels of size, angle, and curvature and the identity channels of shape and motion

Color Vision

The retina of the eye has 2 different kinds of receptors.

- The **rods** actively contribute to vision only in low-light.
- The main sensors in normal lighting conditions are the **cones**.

Color Spaces

- The color space of what colors the human visual system can detect is three dimensional; that is, it can be adequately described using three separate axes.

RGB System

- The most common color space in computer graphics is the system where colors are specified as triples of red, green, and blue values.
- Although this system is computationally convenient, it is a very poor match for the mechanics of how we see.
- The red, green, and blue axes of the RGB color space are not useful as separable channels; they give rise to the integral perception of a color.

HSL System

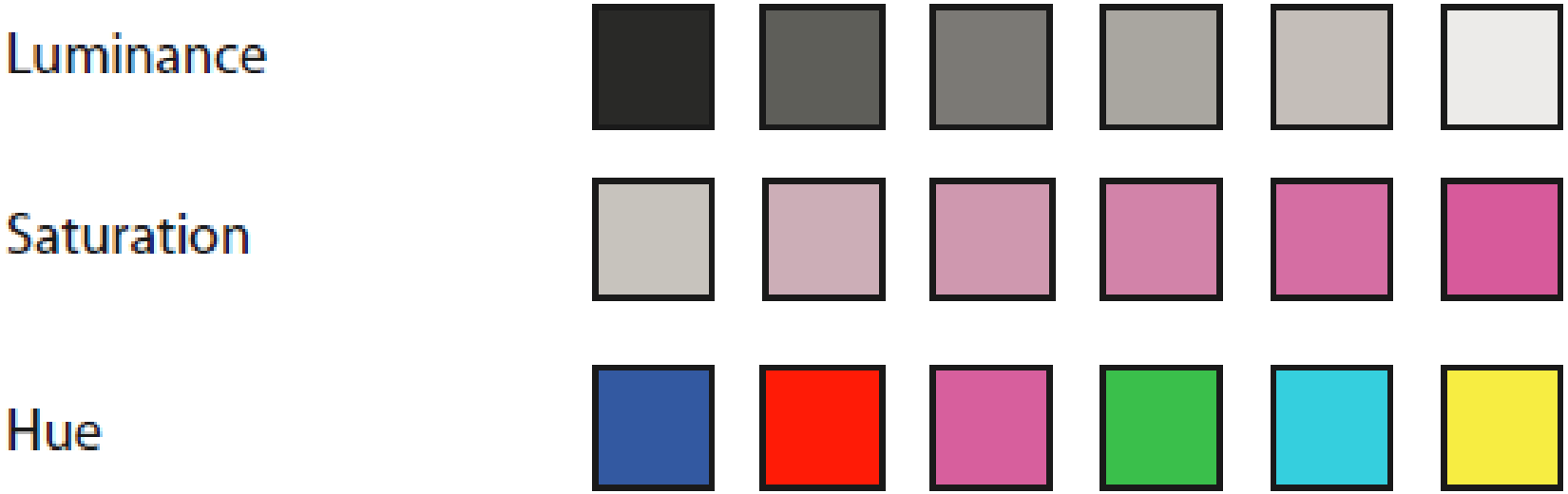
- The hue–saturation–lightness or HSL system is more intuitive and is heavily used by artists and designers.
- The hue axis captures what we normally think of as pure colors that are not mixed with white or black: red, blue, green, yellow, purple, and so on.
- The saturation axis is the amount of white mixed with that pure color. For instance, pink is a partially desaturated red.
- The lightness axis is the amount of black mixed with a color.

HSL System

- The hue–saturation–lightness or HSL system is more intuitive and is heavily used by artists and designers.
- The hue axis captures what we normally think of as pure colors that are not mixed with white or black: red, blue, green, yellow, purple, and so on.
- The saturation axis is the amount of white mixed with that pure color. For instance, pink is a partially desaturated red.
- The lightness axis is the amount of black mixed with a color.

HSL System

- Luminance and saturation are magnitude channels, while hue is a identity channel.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS
WRIGHT STATE UNIVERSITY

Transparency

A fourth channel strongly related to the other three color channels is **transparency**: information can be encoded by decreasing the opacity of a mark from fully opaque to completely see-through.

- Transparency cannot be used independently of the other color channels because of its strong interaction effects with them.
- Transparency is used most often with superimposed layers, to create a foreground layer that is distinguishable from the background layer.
- It is frequently used redundantly, where the same information is encoded with another channel as well.

PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Colormap

A **colormap** specifies a mapping between colors and data values; that is, a visual encoding with color.

Using color to encode data is a powerful and flexible design choice, but colormap design has many pitfalls for the unwary.

Colormaps can be **categorical** or **ordered**, and ordered colormaps can be either **sequential** or **diverging**.

Categorical Colormap

- A **categorical** colormap uses color to encode categories and groupings.
- Categorical colormaps are normally segmented. They are are also known as **qualitative** colormaps.
- Very effective when used appropriately; for categorical data, they are the next best channel after spatial position.
- Categorical colormaps are typically designed by using color as an integral identity channel to encode a single attribute, rather than to encode three completely separate attributes with the three channels of hue, saturation, and luminance.

Categorical Colormap

- An **ordered** colormap is appropriate for encoding ordinal or quantitative attributes.
- A **sequential** colormap ranges from a minimum value to a maximum value.
- A **diverging** colormap has two hues at the endpoints and a neutral color as a midpoint, such as white, gray, or black, or a high-luminance color such as yellow.

Other Channels

Size Channels:

- Size is a magnitude channel suitable for ordered data.
- Length is one-dimensional (1D) size; more specifically, height is vertical size and width is horizontal size. Area is two-dimensional (2D) size, and volume is three-dimensional (3D) size.
- Our judgements of length are extremely accurate.
- Our judgement of area is significantly less accurate.
- The volume channel is quite inaccurate.

Other Channels

Angle Channels

- The *angle* channel encodes magnitude information based on the **orientation** of a mark: the direction that it points.
- There are two slightly different ways to consider orientation that are essentially the same channel. With **angle**, the orientation of one line is judged with respect to another line. With **tilt**, an orientation is judged against the global frame of the display.
- This channel is somewhat less accurate than length and position, it is more accurate than area.

PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by act No. 41 of 2013
40 YEARS OF ACADEMIC WISDOM
WRIGHT STATE UNIVERSITY

Other Channels

Curvature Channel

- The **curvature** channel is not very accurate, and it can only be used with line marks.
- It cannot be used with point marks that have no length, or area marks because their shape is fully constrained.
- The number of distinguishable bins for this channel is low, probably around two or three; it is in an equivalence class with volume (3D size) at the bottom of the magnitude channel ranking.

Other Channels

Shape Channels

- Shape as a identity channel that can be used with point and line marks.
- Applying the shape channel to line marks results in stipple patterns such as dotted and dashed lines.

Motion Channels

- Several kinds of **motion** are also visual channels, including **direction** of motion, **velocity** of motion, and flicker **frequency**.

Heat Map

What is Heat Map?

- A **heat map** is the visualization of data that represents the magnitude of a value in a color code ranging from minor to major intensity.
- The name metaphor is born in the technique used to depict heat; blue means cool, red means hot, and intermediate temperatures are coded as the gradient between those two. The variation of color is often portrayed in intensity and hue, highlighting the extent of the phenomenon to make it easier to interpret.

Types of Heat Map

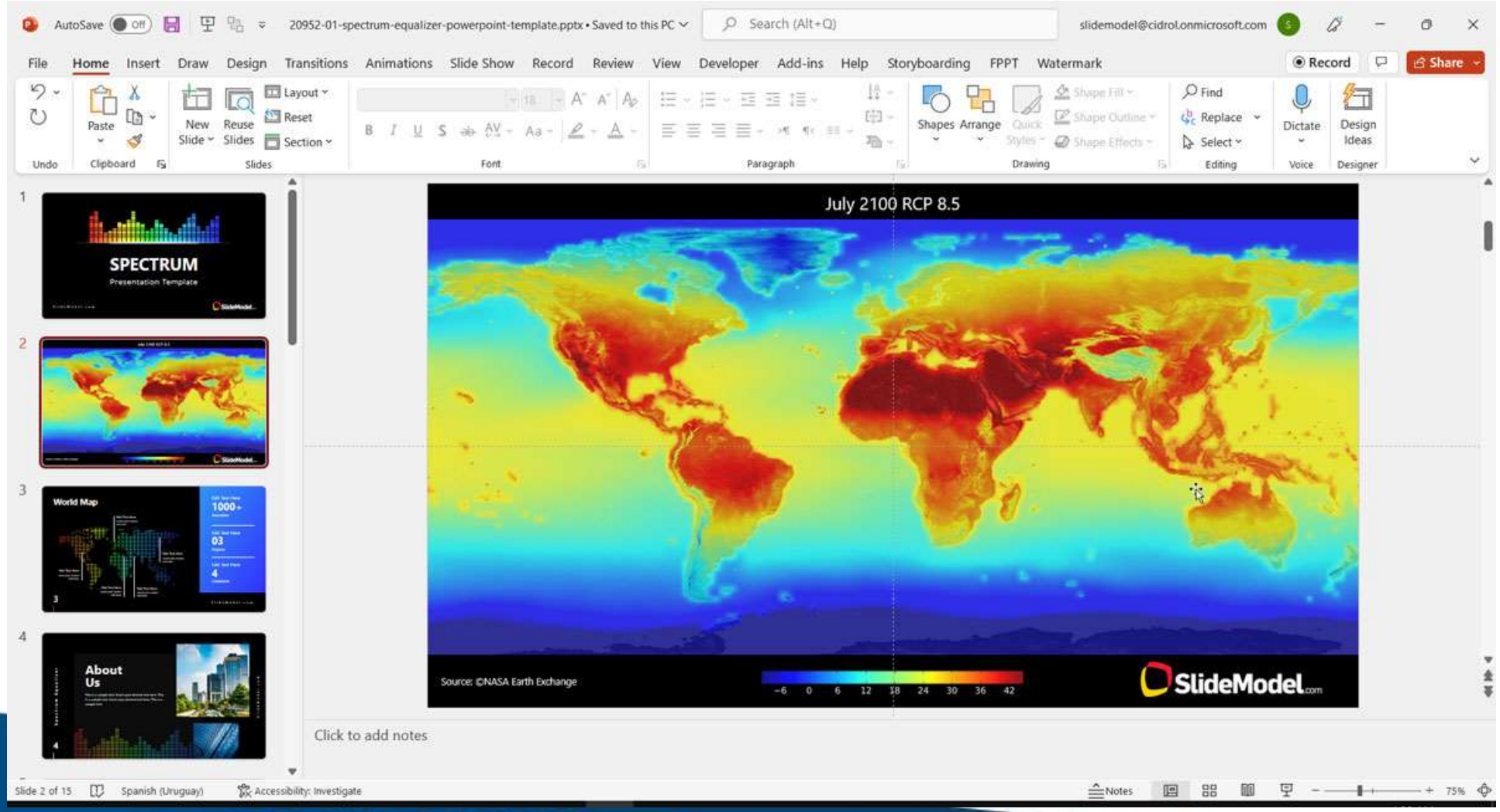
- **The Spatial Heat Map:** Represented with a canvas that represents a two-dimensional space; it can be a geographical map, a web page, or other cartesian representation.
- **Grid Heatmap:** This type of heat map displays the magnitude of a phenomenon using a two-dimensional matrix. Columns and Rows categorize a cell (the location) and the cell's color code defines the phenomenon's value intensity.

Heat map Visualizations

Geographical Heatmap

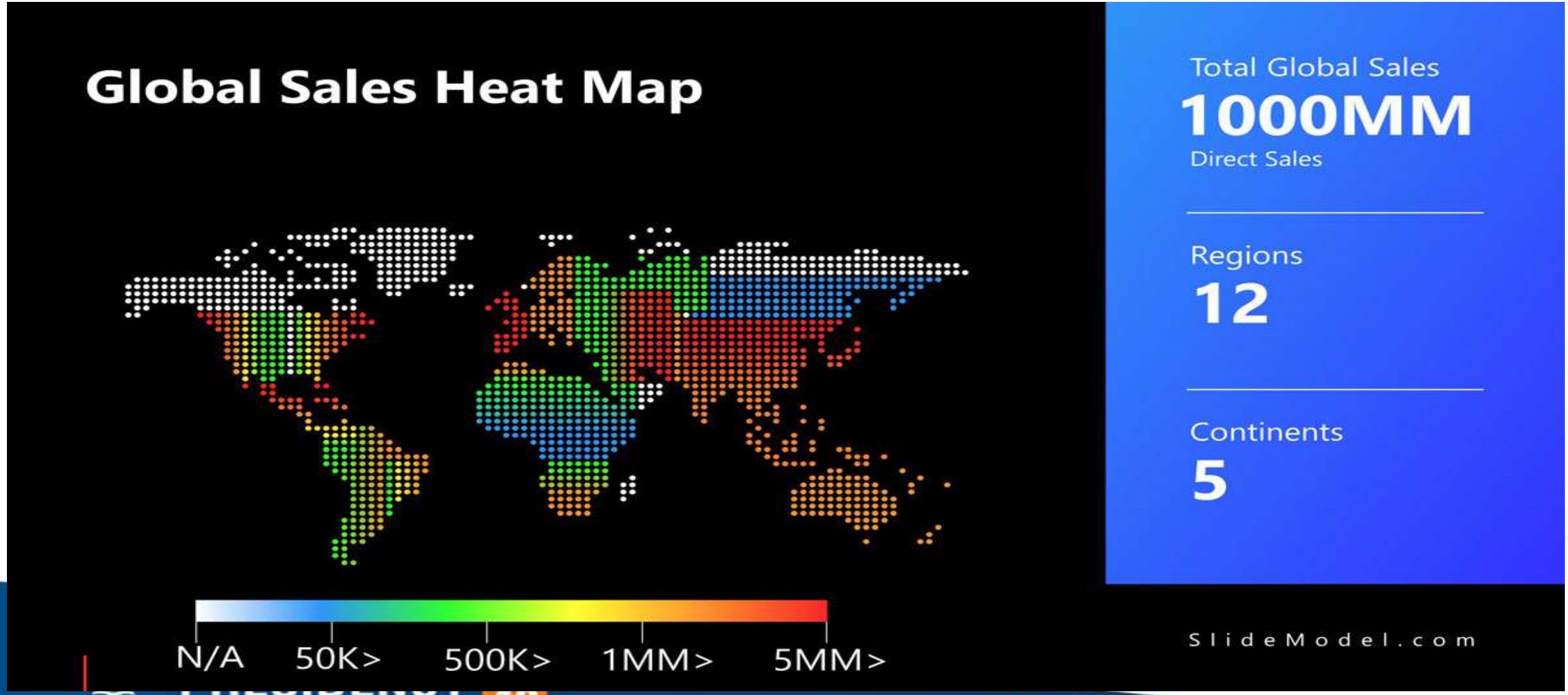
- A geographical heatmap is a spatial map to visualize data according to geographical location. This can be done to show the phenomenon's intensity, such as weather trends or demographic information.
- The heatmap quality will be determined by the density of the dots in the map, and the color spread.
- This means that the more latitude-longitude dots you have on the map, the better the representation of reality.
- Precision geographical heat maps are created with mathematical-statistical tools, like R, Python, or more specialized tools, and require a lot of data.

Heat map Visualizations



Heat Map Representing the temperature of the world during 2100: by NASA

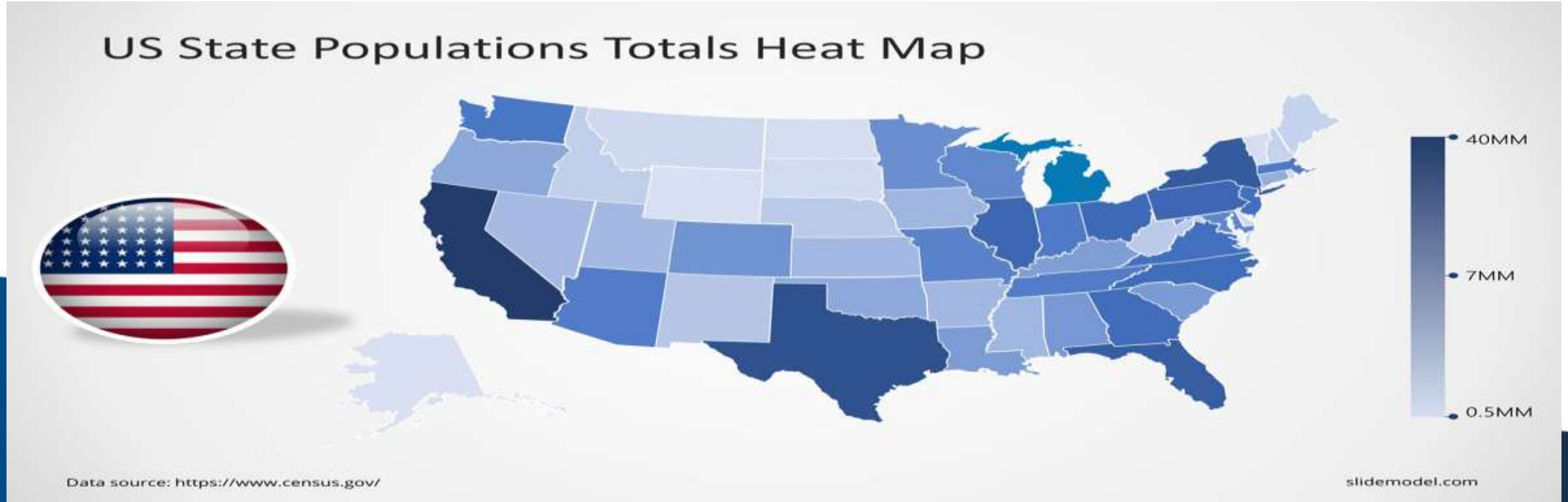
Heat map Visualizations



Heat map Visualizations

Choropleth Maps – An alternative to Geographical Heat Maps

- This differ from the geographic spatial heat maps.
- Mainly because it does not use pure spatial information.
- It uses predefined segments in the map (area/region) to aggregate the phenomenon variable.
- This aggregation makes it simpler to understand and it has become even a more popular tool in business, politics, and social studies.

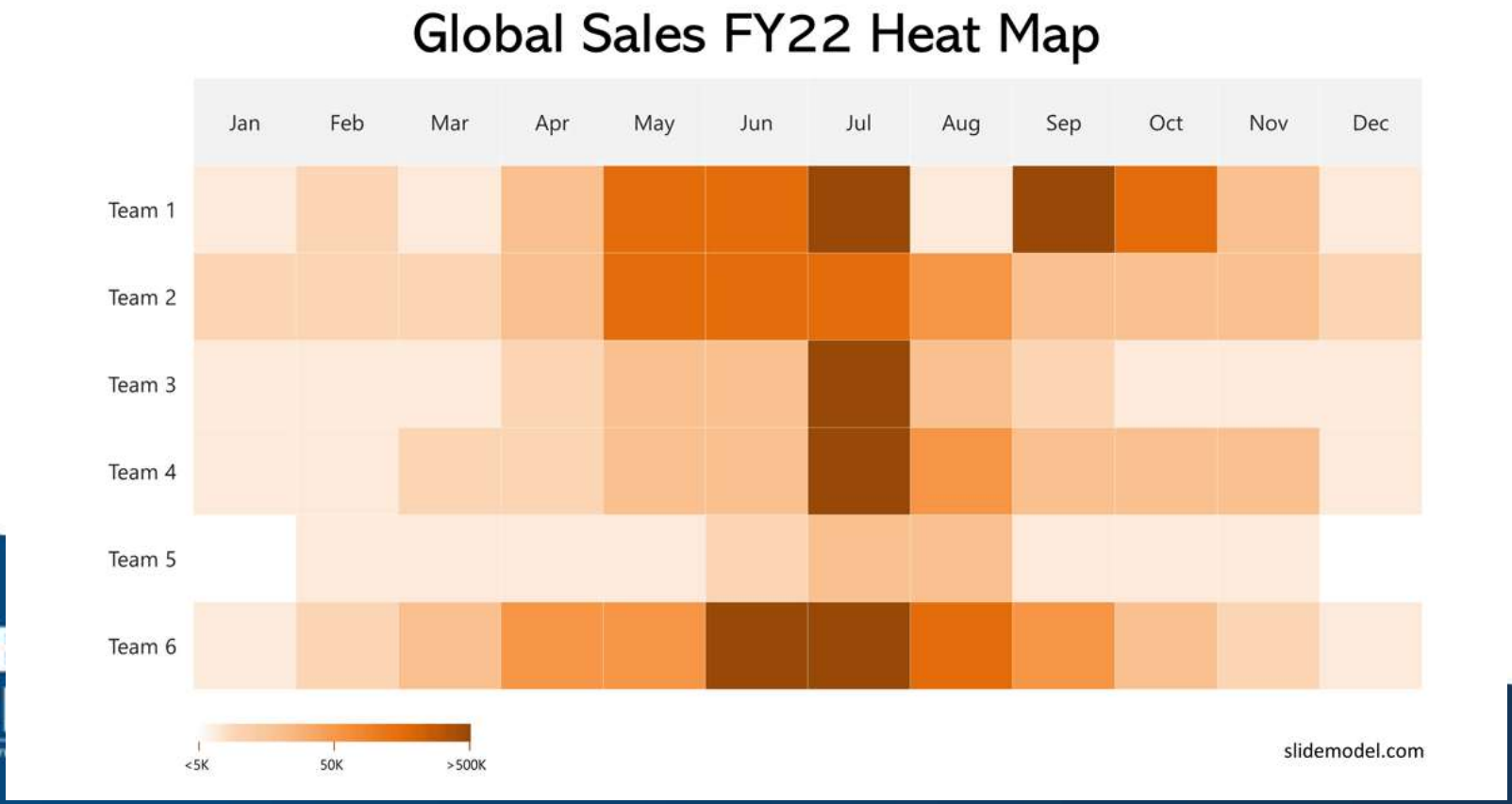


Bubble Chart Heatmap

- Bubble charts are a generalization of the scatter plot.
- Each point is located on a cartesian axis (X, Y) and a circle is created in it.
- The size of the circle is a third dimension in the visualization, used to represent a magnitude.
- When complemented with a color gradient, a fourth dimension can be represented as a heat map.
- Bubble charts can be used both spatially and in the form of a grid.

Matrix Heatmap

- The Matrix heatmap uses a two-dimensional matrix to represent the phenomenon.
- This is basically a grid map with rows, columns, and cell colors to represent data.
- The matrix heatmap depicts the magnitude of a phenomenon based on a 2D matrix, with each category or trait representing a dimension (e.g. year, month, and temperature).
- An example of a matrix heat map is the following analysis of the sales team.



Clustered Heatmap

- Clustered heat maps are a specialization of matrix heat maps; generally used in medicine, biological studies, and mathematics.
- Their purpose is to aid in the visual comparison of sample sets.
- To understand the structure, everything starts with a matrix heat map, where columns are “measuring sets”, and rows are the measured variable.
- Each cell contains the magnitude measured for the pair set-value.
- Then a clustering algorithm is applied to create Dendrograms, first for the variables (rows), later for the sets (columns).

Clustered Heatmap

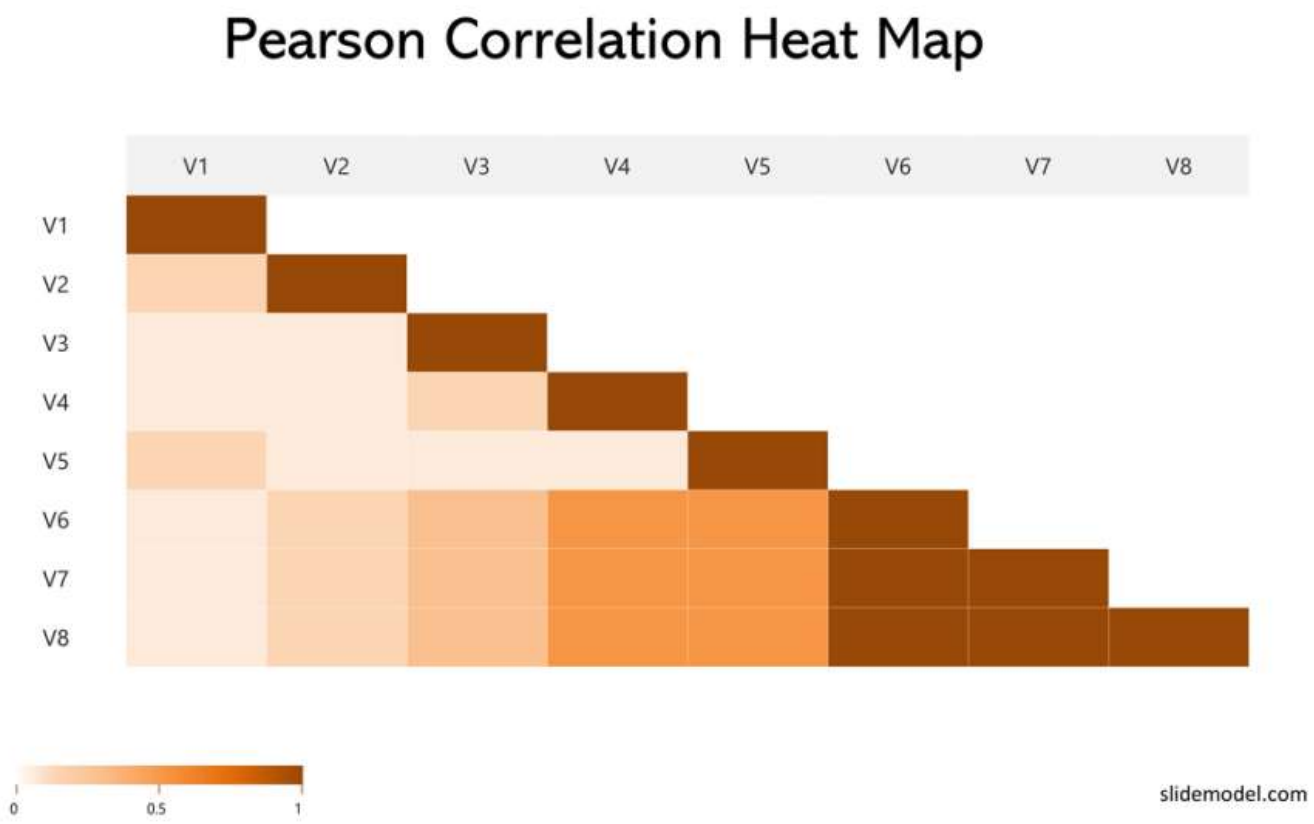
Patients

Measures	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5
DBP (mmHg)	70	76	86	75	81
SBP (mmHg)	130	151	147	120	133
Weight (kg)	100	85	90	75	65
Height (cm)	180	170	165	170	165
Cholesterol (mg/dl)	180	175	220	190	250
Age (years)	35	40	60	35	65

slidemodel.com

Correlation Heatmap

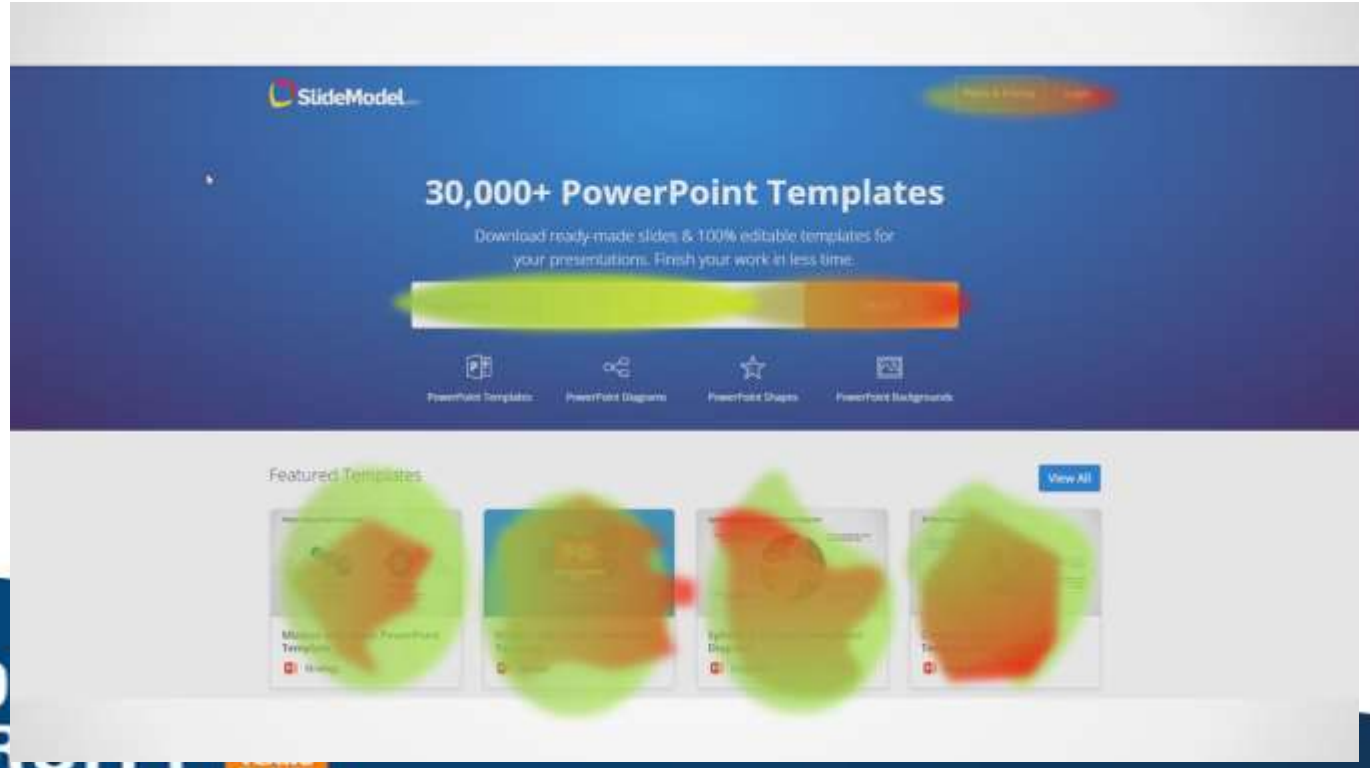
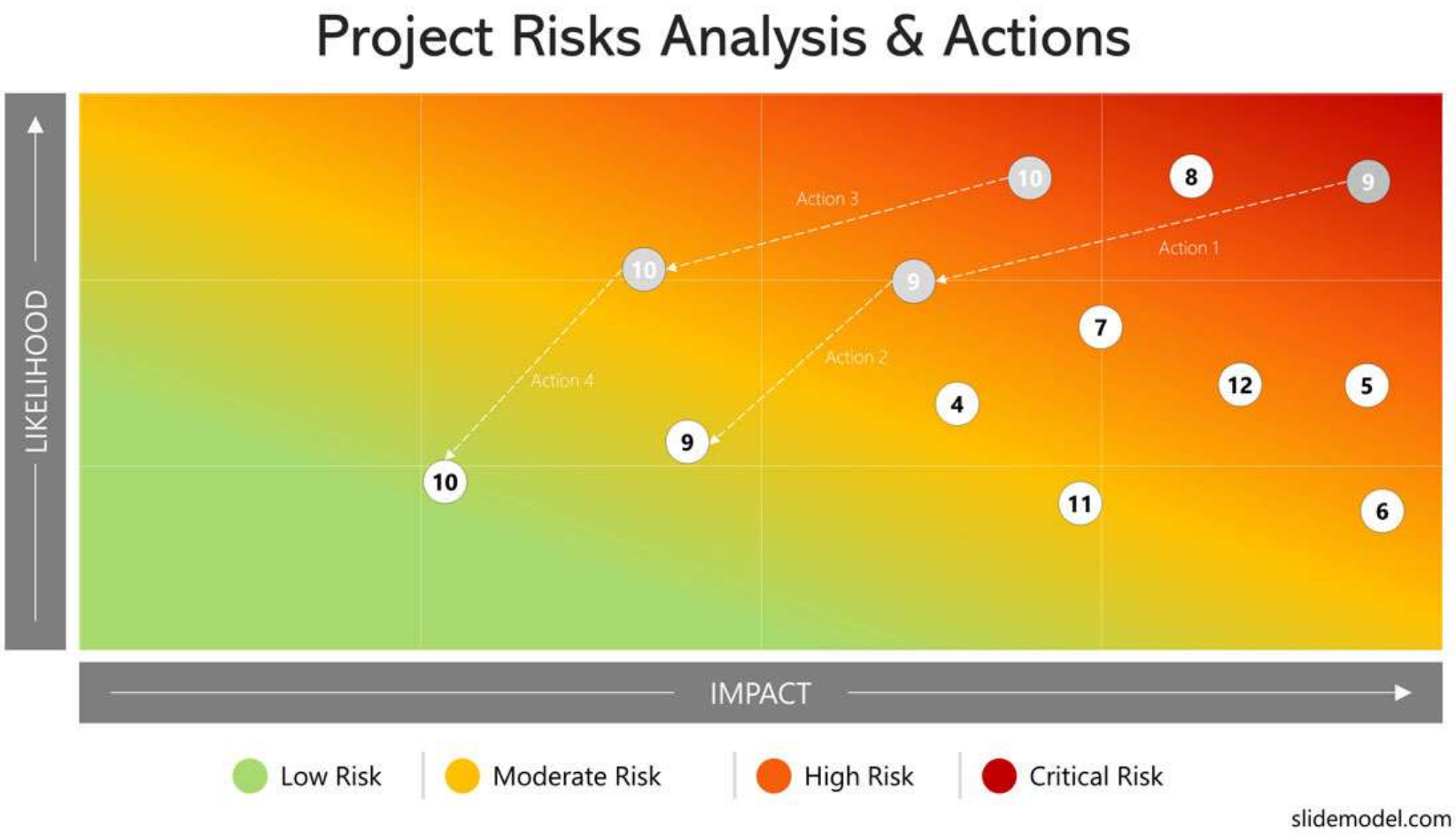
- This visualization is used to interpret the correlation phenomena of a set of measured variables.



PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
40 YEARS

Abstract Positioning Heat Map

- Abstract positioning heatmaps are those where the spatial canvas is not a geographical map, but another kind of plane that wants to be analyzed by a ranging phenomenon.
- The positions in the plane are determined by a cartesian X, Y-axis; the plane is set as a background, and the phenomena are placed with the intensity defined by the color code selected.



Abstract Positioning Heat Map